

Unified Analysis of Decentralized Gradient Descent: a Contraction Mapping Framework

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ABSTRACT The decentralized gradient descent (DGD) algorithm, and its sibling, diffusion, are workhorses in decentralized machine learning, distributed inference and estimation, and multi-agent coordination. We propose a novel, principled framework for the analysis of DGD and diffusion for strongly convex, smooth objectives, and arbitrary undirected topologies, using contraction mappings coupled with a result called the mean Hessian theorem (MHT). The use of these tools yields tight convergence bounds, both in the noise-free and noisy regimes. While these bounds are qualitatively similar to results found in the literature, our approach using contractions together with the MHT decouples the algorithm dynamics (how quickly the algorithm converges to its fixed point) from its asymptotic convergence properties (how far the fixed point is from the global optimum). This yields a simple, intuitive analysis that is accessible to a broader audience. Extensions are provided to multiple local gradient updates, time-varying step sizes, noisy gradients (stochastic DGD and diffusion), communication noise, and random topologies.

INDEX TERMS decentralized machine learning, decentralized optimization, decentralized gradient descent (DGD), diffusion, stochastic DGD and diffusion, federated learning, communication noise, random topology, convergence, contractions, fixed points, mean Hessian theorem

I. Introduction

In applications such as decentralized machine learning [1], distributed inference and estimation [2], [3], and multi-agent coordination [4], N agents interconnected in a communication network cooperate on the minimization of a global objective function $F(x) = \sum_{n=1}^N f_n(x)/N$, where $f_n(x)$ represents the local objective function at the n th agent. The algorithmic workhorse for minimizing $F(x)$ is decentralized gradient descent (DGD), and its noisy (stochastic) versions.

This paper presents a textbook-style, from-first-principles analysis of DGD and its variants for strongly convex, smooth objectives, and arbitrary, undirected network topologies. Our focus on strongly convex and smooth optimization problems is motivated by their stronger convergence guarantees, simplified analysis, and theoretical insights. In fact, the insights gained from their analysis can often guide the design of novel algorithms in more complex settings, serving as a cornerstone for both theoretical and practical developments.

Our main analytical tools are the mean Hessian theorem (MHT), and a succession of theorems that deal with contraction mappings: the contraction convergence theorem (C^2T), the concatenated C^2T (C^3T), and the noisy C^3T (NC^3T). We first use these contraction theorems together with the MHT to establish a tight contraction to a fixed point, and then, independently, use the MHT to find a tight bound on the distance from this fixed point to the optimal solution. Via the MHT, the difference between the gradients at different points x and y can be expressed as a linear function of $y - x$, akin to a quadratic function: $\nabla f(y) - \nabla f(x) = A_{xy}(y - x)$, where A_{xy} is a positive definite matrix (the mean Hessian). The MHT allows leveraging properties of positive definite matrices in the analysis of DGD algorithms, significantly simplifying their analysis. In particular, we show that the MHT encapsulates all key properties of strong convexity and smoothness, which are typically used separately in convergence proofs, within a single linear equation.

Beyond the basic DGD algorithm, we cover a number of extensions, including: the diffusion version of DGD, multiple local gradient updates, time-varying step sizes, sampling noise (stochastic DGD), communication noise, random topologies, and link failures.

A. Literature on DGD and its Variants

The core ideas behind the DGD algorithm go back at least to [5]. The literature on its analysis is quite large. The analysis in [6] uses an approach with subgradients to handle objectives that are convex but not necessarily smooth, and connectivity graphs that vary over time. A key assumption is that the subgradients are uniformly bounded and that the time-variation of the connectivity satisfies certain conditions. In [7], an analysis is given for smooth and convex, but not necessarily strongly convex, functions. Several variations of the basic DGD algorithm have been developed. An important example is the diffusion algorithm [8]. Other extensions of DGD are gradient tracking [9]–[11] and DGD for optimization over directed topologies [12].

A special case is when the communication network topology is a complete graph. In this case, popularized as *federated learning* in [13], in each iteration every agent exchanges information with everyone else, typically via a central server.¹ DGD (or diffusion) for the complete-graph scenario is sometimes called *distributed* gradient descent, and its analysis follows as a special case.

Our exposition targets a broad audience, providing an accessible exposition of the main principles and an entry point to the field. The uniqueness of our treatment lies in the MHT+contraction framework that allows a separate analysis of the convergence to a fixed point, and the distance from this fixed point to the optimal point. Among other papers with unified analyses of decentralized optimization algorithms we note especially [14]–[16] – these papers contain a comprehensive treatment of the convergence rate for many different algorithms under a rich variety of assumptions, tailored to a more specialized readership. Noteworthy are also the overview papers [17], [18], and the book [19], with tutorial-style expositions of different versions of DGD.

In the last few years, a sizeable body of literature has appeared on federated and fully decentralized learning over random networks [20], communication-constrained and unreliable networks [21]–[26], and wireless networks [27]–[36]. Good overview papers of this literature are [37]–[41]. Our MHT+contraction framework can naturally handle the impact of communication noise, randomness of the topology, and link failures.

¹The interconnection topology is a star (with the server at the center), but since the only role of the server is to perform aggregation, the effective topology for the decentralized optimization becomes a complete graph.

B. Specific Technical Contributions

Our main novel contribution is the MHT+contraction framework, enabling a unified analysis of DGD and diffusion. In addition, we make the following specific contributions:

- 1) The statement and proof of the NC³T theorem for noisy contractions.
While the C²T and C³T are immediate consequences of standard fixed point theorems, NC³T is a new and powerful result that makes a clean break with previous approaches. The NC³T specifically enables the analysis of noisy contraction mappings for which the noise variance scales with the argument of the mapping.
- 2) The statement and proof of the MHT for only *once differentiable* functions.
The MHT for *twice differentiable* functions is well known, see the proof of Lemma 1.2.2 in [42], and was used before to analyze DGD – for example in [43]–[45].
- 3) A discussion of tightness of our convergence bounds for DGD and diffusion, showing that the contraction factors are the best possible.
- 4) A discussion of various special cases, including that of bipartite topologies, and of various misconceptions.

The limitations of our framework are: (1) the objectives must be (once) differentiable, strongly convex, and smooth; and (2) the weight matrix must be symmetric. Nonetheless, when treating random topologies (Section V.C), it is sufficient that the weight matrix is symmetric *in expectation*; individual realizations of it may be asymmetric and directed.

C. Roadmap

Section II gives the main mathematical tools, the contraction theorems (C²T, C³T, and NC³T), and the MHT. Section III shows, as a preparation for the analysis of the DGD and diffusion algorithms, how the MHT+contraction framework can be used to analyze gradient descent (GD). Section IV contains the main analysis of DGD and diffusion, for single and multiple local gradient updates, and for time-varying step sizes. Section V extends the analyses to the cases of gradient sampling noise, communication noise, and random topologies. Section VI exemplifies how these analyses apply to DGD with random link failures. Section VII provides additional discussion and insights, and extensions. Section VIII concludes the paper.

D. Notation and Conventions

Lowercase bold symbols are column vectors, and uppercase bold symbols are matrices. $\mathbf{0}$ is the all-zeros vector/matrix. $\mathbf{I}$ is the identity matrix, and $\{\mathbf{e}_n\}$ denote the columns of $\mathbf{I}$. $(\cdot)^T$ denotes the transpose of a vector or matrix. $\mathbf{1}$ is the vector of all ones: $\mathbf{1} = [1, \dots, 1]^T$. For a vector $\mathbf{x}$, $\|\mathbf{x}\|$ is its Euclidean norm. For a matrix $\mathbf{X}$, $\mathbf{X}^{-1}$ is its inverse. $\text{vec}\{\mathbf{X}\}$ denotes the vector obtained by stacking the columns of a matrix $\mathbf{X}$ on top of one another, and $\otimes$ denotes the Kronecker product. $\Pi_{\mathbf{X}} = \mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T$ denotes the orthogonal projection onto the column space of $\mathbf{X}$, and $\Pi_{\mathbf{X}}^\perp = \mathbf{I} - \Pi_{\mathbf{X}}$ denotes

the orthogonal projection onto the orthogonal complement of this space. $\mathbf{X}^\dagger$ denotes the Moore-Penrose pseudoinverse. For symmetric matrices $\mathbf{X}$ and $\mathbf{Y}$ of compatible dimensions, $\mathbf{X} \succcurlyeq \mathbf{Y}$ means that $\mathbf{X} - \mathbf{Y}$ is positive semidefinite. $\|\mathbf{X}\|_2$ is the spectral norm of the matrix $\mathbf{X}$. ∇ and ∇^2 denote the gradient and Hessian operators, respectively; as a special case, $(\cdot)'$ denotes derivative. $\mathcal{O}(x)$ stands for big ordo and $\mathcal{O}(x)$ denotes a matrix whose spectral norm is $\mathcal{O}(x)$. $(\cdot)^+$ stands for $\max(0, \cdot)$.

Explicit conditions on η (to be used for the step size) will be stated when their dependence on the problem offers significant insight. At other places, to avoid digression into uninteresting calculations, we say that an expression is *valid for small enough* η if there exists η' such that the expression holds for $\eta \leq \eta'$, without giving η' explicitly. For example, $\eta/\sqrt{2\eta - \eta^2} \leq \sqrt{\eta}$ for small enough η .

II. Mathematical Tools

A. Contractions and Fixed Points

A function $\phi: \mathbb{R}^m \rightarrow \mathbb{R}^m$ is called a *contraction* (on $\mathbb{R}^m$) if there exists a constant c , $c < 1$, such that

$$\|\phi(\mathbf{y}) - \phi(\mathbf{x})\| \leq c\|\mathbf{y} - \mathbf{x}\| \quad (1)$$

for all $\mathbf{x}$ and $\mathbf{y}$. For a contraction $\phi(\cdot)$, Banach's fixed point theorem [46, Th. 9.23] states that there is a unique *fixed point*, $\hat{\mathbf{x}}$, such that

$$\phi(\hat{\mathbf{x}}) = \hat{\mathbf{x}}.$$

The following consequence of the fixed point theorem will be one of our main tools.

Contraction Convergence Theorem (C²T): Let $\phi(\cdot)$ be a contraction as defined in (1), and consider the recursion

$$\mathbf{x}_{t+1} = \phi(\mathbf{x}_t)$$

for $t = 0, 1, 2, \dots$ and some arbitrary $\mathbf{x}_0$. Then

$$\|\mathbf{x}_t - \hat{\mathbf{x}}\| \leq c^t \|\mathbf{x}_0 - \hat{\mathbf{x}}\|.$$

In particular, $\mathbf{x}_t \rightarrow \hat{\mathbf{x}}$ as $t \rightarrow \infty$. ■

The C²T follows immediately from the fact that

$$\|\mathbf{x}_{t+1} - \hat{\mathbf{x}}\| = \|\phi(\mathbf{x}_t) - \phi(\hat{\mathbf{x}})\| \leq c\|\mathbf{x}_t - \hat{\mathbf{x}}\|,$$

together with induction over t .

A mapping formed by nesting contractions is also a contraction. More formally we have the following.

Concatenated Contraction Convergence Theorem (C³T): Let $\{\phi_1(\cdot), \dots, \phi_n(\cdot)\}$ be contractions, with contraction factors $\{c_1, \dots, c_n\}$, respectively. Then the concatenated mapping

$$\varphi(\mathbf{x}) = \phi_n(\dots(\phi_1(\mathbf{x}))) \quad (2)$$

is also a contraction. Moreover, the recursion

$$\mathbf{x}_{t+1} = \varphi(\mathbf{x}_t)$$

has the property

$$\|\mathbf{x}_t - \hat{\mathbf{x}}\| \leq \left(\prod_{i=1}^n c_i \right)^t \|\mathbf{x}_0 - \hat{\mathbf{x}}\|, \quad (3)$$

where $\hat{\mathbf{x}}$ is the fixed point of $\varphi(\cdot)$. ■

Proof of the C³T

Recursive application of (1) shows that $\varphi(\cdot)$ is a contraction:

$$\begin{aligned} \|\varphi(\mathbf{y}) - \varphi(\mathbf{x})\| &= \|\phi_n(\dots(\phi_1(\mathbf{y}))) - \phi_n(\dots(\phi_1(\mathbf{x})))\| \\ &\leq c_n \|\phi_{n-1}(\dots(\phi_1(\mathbf{y}))) - \phi_{n-1}(\dots(\phi_1(\mathbf{x})))\| \\ &\leq c_n c_{n-1} \|\phi_{n-2}(\dots(\phi_1(\mathbf{y}))) - \phi_{n-2}(\dots(\phi_1(\mathbf{x})))\| \\ &\leq \dots \leq c_n c_{n-1} \dots c_2 \|\phi_1(\mathbf{y}) - \phi_1(\mathbf{x})\| \\ &\leq \left(\prod_{i=1}^n c_i \right) \|\mathbf{y} - \mathbf{x}\|. \end{aligned}$$

Banach's fixed point theorem guarantees the existence of $\hat{\mathbf{x}}$ such that $\varphi(\hat{\mathbf{x}}) = \hat{\mathbf{x}}$. Then (3) follows from the C²T applied to $\varphi(\cdot)$. ■

B. Noisy Contractions

If the recursions in the C²T or C³T contain noise, then under mild conditions convergence holds in the mean-square sense. We give a noisy version of the C³T, from which a noisy version of the C²T follows as a special case.

Noisy C³T (NC³T): Let $\{\phi_1(\cdot), \dots, \phi_n(\cdot)\}$ be contractions, each with the same contraction factor c . Define a (noisy) mapping from $\mathbf{x}_t$ onto $\mathbf{x}_{t+1}$ via the noisy recursions

$$\begin{aligned} \mathbf{x}_{t,0} &= \mathbf{x}_t \\ \mathbf{x}_{t,1} &= \phi_1(\mathbf{x}_{t,0}) + \boldsymbol{\epsilon}_{t,0} \\ \mathbf{x}_{t,2} &= \phi_2(\mathbf{x}_{t,1}) + \boldsymbol{\epsilon}_{t,1} \\ &\dots \\ \mathbf{x}_{t,n} &= \phi_n(\mathbf{x}_{t,n-1}) + \boldsymbol{\epsilon}_{t,n-1} \\ \mathbf{x}_{t+1} &= \mathbf{x}_{t,n}, \end{aligned}$$

where $\{\boldsymbol{\epsilon}_{t,i}\}$ are noise terms that satisfy

$$\mathbb{E}\{\boldsymbol{\epsilon}_{t,i} | \mathbf{x}_{t,i}\} = \mathbf{0}, \quad (4)$$

$$\sqrt{\mathbb{E}\{\|\boldsymbol{\epsilon}_{t,i}\|^2 | \mathbf{x}_{t,i}\}} \leq \sigma + \omega \|\mathbf{x}_{t,i}\|, \quad (5)$$

for some non-negative constants σ and ω , and suppose

$$c^2 + \omega^2 < 1. \quad (6)$$

Let the contraction $\varphi(\cdot)$ be defined as in (2), and let $\hat{\mathbf{x}}$ be its fixed point. Then

$$\sqrt{\mathbb{E}\{\|\mathbf{x}_t - \hat{\mathbf{x}}\|^2\}} \leq \hat{d} + \sqrt{c^2 + \omega^2}^{nt} \left(\|\mathbf{x}_0 - \hat{\mathbf{x}}\| - \hat{d} \right)^+, \quad (7)$$

where $\hat{d}$ and M are constants given by

$$\hat{d} = \frac{\omega M + \sigma}{\sqrt{1 - c^2} - \omega}, \quad (8)$$

$$M = \max\{\|\tilde{\mathbf{x}}_1\|, \dots, \|\tilde{\mathbf{x}}_n\|\},$$

and where $\{\tilde{\mathbf{x}}_i\}$ constitute transformations of the fixed point through the n intermediate mappings $\phi_1(\cdot), \dots, \phi_n(\cdot)$:

$$\tilde{\mathbf{x}}_1 = \phi_1(\hat{\mathbf{x}})$$

$$\tilde{\mathbf{x}}_2 = \phi_2(\tilde{\mathbf{x}}_1)$$

$$\dots$$

$$\tilde{\mathbf{x}}_n = \phi_n(\tilde{\mathbf{x}}_{n-1}). \quad \blacksquare$$

The NC³T shows that, in the limit $t \rightarrow \infty$, the root-mean square value of $\|\mathbf{x}_t - \hat{\mathbf{x}}\|$ is no larger than $\hat{d}$.

1) Proof of the NC³T

First define the following shorthands:

$$\nu = \sqrt{c^2 + \omega^2},$$

$$\sigma' = \omega M + \sigma.$$

Consider the following quadratic equation:

$$\hat{d}^2 = \nu^2 \hat{d}^2 + 2\sigma' \omega \hat{d} + \sigma'^2. \quad (9)$$

Equation (9) has the unique non-negative solution stated in (8):

$$\begin{aligned} \hat{d} &= \frac{\sigma' \omega}{1 - \nu^2} + \sqrt{\frac{\sigma'^2 \omega^2}{(1 - \nu^2)^2} + \frac{\sigma'^2}{1 - \nu^2}} \\ &= \sigma' \cdot \frac{\omega + \sqrt{1 - c^2}}{1 - c^2 - \omega^2} \\ &= \sigma' \cdot \frac{\omega + \sqrt{1 - c^2}}{(\sqrt{1 - c^2} - \omega)(\sqrt{1 - c^2} + \omega)} \\ &= \frac{\sigma'}{\sqrt{1 - c^2} - \omega} = \frac{\omega M + \sigma}{\sqrt{1 - c^2} - \omega}. \end{aligned}$$

Now define the error,

$$d_{t,i} = \sqrt{\mathbb{E}\{\|\mathbf{x}_{t,i} - \tilde{\mathbf{x}}_i\|^2\}},$$

and consider its evolution with t and i . Let $\tilde{\mathbf{x}}_0 = \hat{\mathbf{x}}$. For given t , we have for $i = 0, \dots, n - 1$ that

$$\begin{aligned} d_{t,i+1}^2 &= \mathbb{E}\{\|\mathbf{x}_{t,i+1} - \tilde{\mathbf{x}}_{i+1}\|^2\} \\ &= \mathbb{E}\{\|\phi_{i+1}(\mathbf{x}_{t,i}) - \phi_{i+1}(\tilde{\mathbf{x}}_i) + \boldsymbol{\epsilon}_{t,i}\|^2\} \\ &= \mathbb{E}\{\|\phi_{i+1}(\mathbf{x}_{t,i}) - \phi_{i+1}(\tilde{\mathbf{x}}_i)\|^2\} + \mathbb{E}\{\|\boldsymbol{\epsilon}_{t,i}\|^2\} \\ &\quad + 2\mathbb{E}\{(\phi_{i+1}(\mathbf{x}_{t,i}) - \phi_{i+1}(\tilde{\mathbf{x}}_i))^T \boldsymbol{\epsilon}_{t,i}\} \\ &\stackrel{(a)}{=} \mathbb{E}\{\|\phi_{i+1}(\mathbf{x}_{t,i}) - \phi_{i+1}(\tilde{\mathbf{x}}_i)\|^2\} + \mathbb{E}\{\|\boldsymbol{\epsilon}_{t,i}\|^2\} \\ &\stackrel{(b)}{\leq} c^2 \mathbb{E}\{\|\mathbf{x}_{t,i} - \tilde{\mathbf{x}}_i\|^2\} + \mathbb{E}\{\|\boldsymbol{\epsilon}_{t,i}\|^2\} \\ &\stackrel{(c)}{\leq} c^2 \mathbb{E}\{\|\mathbf{x}_{t,i} - \tilde{\mathbf{x}}_i\|^2\} + (\sigma' + \omega d_{t,i})^2 \\ &= c^2 d_{t,i}^2 + (\sigma' + \omega d_{t,i})^2 \\ &= \nu^2 d_{t,i}^2 + 2\sigma' \omega d_{t,i} + \sigma'^2, \end{aligned} \quad (10)$$

where in (a) we used (4), in (b) we used (1) and in (c) we used the following:

$$\begin{aligned} \mathbb{E}\{\|\boldsymbol{\epsilon}_{t,i}\|^2\} &= \mathbb{E}\{\mathbb{E}\{\|\boldsymbol{\epsilon}_{t,i}\|^2 | \mathbf{x}_{t,i}\}\} \\ &\stackrel{(a)}{\leq} \mathbb{E}\{(\sigma + \omega \|\mathbf{x}_{t,i}\|)^2\} \\ &= \mathbb{E}\{(\sigma + \omega \|\mathbf{x}_{t,i} - \tilde{\mathbf{x}}_i + \tilde{\mathbf{x}}_i\|)^2\} \\ &\leq \mathbb{E}\{(\sigma + \omega \|\mathbf{x}_{t,i} - \tilde{\mathbf{x}}_i\| + \omega \|\tilde{\mathbf{x}}_i\|)^2\} \\ &\leq \mathbb{E}\{(\sigma' + \omega \|\mathbf{x}_{t,i} - \tilde{\mathbf{x}}_i\|)^2\} \\ &= \sigma'^2 + \omega^2 \mathbb{E}\{\|\mathbf{x}_{t,i} - \tilde{\mathbf{x}}_i\|^2\} + 2\sigma' \omega \mathbb{E}\{\|\mathbf{x}_{t,i} - \tilde{\mathbf{x}}_i\|\} \\ &\stackrel{(b)}{\leq} \sigma'^2 + \omega^2 \mathbb{E}\{\|\mathbf{x}_{t,i} - \tilde{\mathbf{x}}_i\|^2\} + 2\sigma' \omega \sqrt{\mathbb{E}\{\|\mathbf{x}_{t,i} - \tilde{\mathbf{x}}_i\|^2\}} \\ &= (\sigma' + \omega d_{t,i})^2. \end{aligned} \quad (11)$$

In (11), (a) follows from (5) and (b) follows from Jensen's inequality and the concavity of $\sqrt{\cdot}$.

Consider (10). If $d_{t,i} \leq \hat{d}$, then

$$\begin{aligned} d_{t,i+1} &\leq \sqrt{\nu^2 d_{t,i}^2 + 2\sigma' \omega d_{t,i} + \sigma'^2} \\ &\leq \sqrt{\nu^2 \hat{d}^2 + 2\sigma' \omega \hat{d} + \sigma'^2} = \hat{d}. \end{aligned}$$

If $d_{t,i} \geq \hat{d}$, then

$$\begin{aligned} d_{t,i+1} &\leq \nu d_{t,i} + \left(\sqrt{\nu^2 d_{t,i}^2 + 2\sigma' \omega d_{t,i} + \sigma'^2} - \nu d_{t,i} \right) \\ &\stackrel{(a)}{\leq} \nu d_{t,i} + \left(\sqrt{\nu^2 \hat{d}^2 + 2\sigma' \omega \hat{d} + \sigma'^2} - \nu \hat{d} \right) \\ &= \nu d_{t,i} + \hat{d} - \nu \hat{d} \\ &= \hat{d} + \nu(d_{t,i} - \hat{d}), \end{aligned}$$

where in (a) we used that $\sqrt{\nu^2 d^2 + 2\sigma' \omega d + \sigma'^2} - \nu d$ is a decreasing function of d ; this is so because its derivative,

$$\frac{\nu^2 d + \sigma' \omega}{\sqrt{\nu^2 d^2 + 2\sigma' \omega d + \sigma'^2}} - \nu,$$

is non-positive if $(\nu^2 d + \sigma' \omega)^2 \leq \nu^2 (\nu^2 d^2 + 2\sigma' \omega d + \sigma'^2)$, which always holds.

Induction over i gives that

$$d_{t,n} \leq \hat{d} + \nu^n (d_{t,0} - \hat{d})^+. \quad (12)$$

Since $\hat{\mathbf{x}}$ is the fixed point of $\varphi(\cdot)$, we have $\tilde{\mathbf{x}}_n = \hat{\mathbf{x}}$, which implies that

$$d_{t+1,0} = d_{t,n}. \quad (13)$$

Using (12) and (13) with induction over t , (7) follows. $\blacksquare$

2) Remarks

- In the special case of $n = 1$, the NC³T simplifies to a noisy version of the C²T. Suppressing superfluous subscripts, the recursion is

$$\mathbf{x}_{t+1} = \phi(\mathbf{x}_t) + \boldsymbol{\epsilon}_t,$$

$\tilde{\mathbf{x}}_1 = \hat{\mathbf{x}}$ is the fixed point of $\phi(\cdot)$, and $M = \|\hat{\mathbf{x}}\|$.

We refer to this special case as the noisy C²T (NC²T).

- In the noise-free case, $\sigma = \omega = 0$, so $\hat{d} = 0$ and $\sqrt{\mathbb{E}\{\|\mathbf{x}_t - \hat{\mathbf{x}}\|^2\}} = \|\mathbf{x}_t - \hat{\mathbf{x}}\|$. Then (7) reduces to (3), so the C³T follows as a special case of the NC³T.
- If $\{\phi_i(\cdot)\}$ are contractions with *different* factors $\{c_i\}$, then the NC³T holds with $c = \max\{c_1, \dots, c_n\}$.
- When $\omega > 0$, the conditional noise variance scales with $\|\mathbf{x}_{t,i}\|^2$. This slows down the convergence, increasing the contraction factor from c to $\sqrt{c^2 + \omega^2}$. In fact, if $c^2 + \omega^2 \geq 1$, the mean-square error may diverge even if the noise-free recursion is a contraction. For example, consider a scalar x , take $n = 1$, and $\phi_1(x) = x/2$. Suppose ϵ_t is selected such that $\mathbb{E}\{\epsilon_t|x\} = 0$ and $\mathbb{E}\{\epsilon_t^2|x\} = (1 + \sqrt{3}/2|x|)^2$. Then with $\sigma = 1$ and $\omega = \sqrt{3}/2$, (4)–(5) are satisfied with equality. It follows that $c = 1/2$, $\hat{x} = 0$, and $c^2 + \omega^2 = 1$, so equality holds in (b) in (10) and $d_{t,1}$ grows without bound:

$$\begin{aligned} d_{t+1,1}^2 &\geq c^2 d_{t+1,0}^2 + \sigma^2 + \omega^2 d_{t+1,0}^2 \\ &= d_{t+1,0}^2 + 1 = d_{t,1}^2 + 1. \end{aligned}$$

Hence, strict inequality in (6) is crucial.

- Suppose (5) holds nominally and that the coordinate system is shifted by adding a constant vector $\mathbf{a}$ to all $\{\mathbf{x}_t\}$ and $\{\mathbf{x}_{t,i}\}$. Since

$$\sigma + \omega\|\mathbf{x}_{t,i} + \mathbf{a}\| \leq \sigma + \omega\|\mathbf{x}_{t,i}\| + \omega\|\mathbf{a}\|,$$

it follows that (5) holds in the new coordinate system with ω unchanged, and σ replaced with $\sigma + \omega\|\mathbf{a}\|$. Importantly, the contraction speed is unaffected.

- Note that

$$\sqrt{\sigma^2 + \omega^2\|\mathbf{x}_{t,i}\|^2} \leq \sigma + \omega\|\mathbf{x}_{t,i}\|.$$

Hence, the following alternative condition on the noise variance (used, for instance, in [45]),

$$\mathbb{E}\{\|\epsilon_t\|^2|\mathbf{x}_{t,i}\}\} \leq \sigma^2 + \omega^2\|\mathbf{x}_{t,i}\|^2 \quad (14)$$

implies (5) in the NC³T and hence is more restrictive.

C. Proving Convergence to a Desired Goal

For iterative algorithms whose updates are contraction mappings, we can use the C²T, C³T, and NC³T to bound the distance between $\mathbf{x}_t$ and a desired goal $\mathbf{x}_*$ (typically, the minimizer of a global objective function). In the noise-free case,

$$\begin{aligned} \|\mathbf{x}_t - \mathbf{x}_*\| &= \|\mathbf{x}_t - \hat{\mathbf{x}} + \hat{\mathbf{x}} - \mathbf{x}_*\| \\ &\stackrel{(a)}{\leq} \|\mathbf{x}_t - \hat{\mathbf{x}}\| + \|\hat{\mathbf{x}} - \mathbf{x}_*\| \\ &\stackrel{(b)}{\leq} c^{nt}\|\mathbf{x}_0 - \hat{\mathbf{x}}\| + \|\hat{\mathbf{x}} - \mathbf{x}_*\|, \end{aligned} \quad (15)$$

where (a) follows from the triangle inequality, and (b) follows from the C²T or C³T. In the noisy case,

$$\begin{aligned} \sqrt{\mathbb{E}\{\|\mathbf{x}_t - \mathbf{x}_*\|^2\}} &= \sqrt{\mathbb{E}\{\|\mathbf{x}_t - \hat{\mathbf{x}} + \hat{\mathbf{x}} - \mathbf{x}_*\|^2\}} \\ &\stackrel{(a)}{\leq} \sqrt{\mathbb{E}\{\|\mathbf{x}_t - \hat{\mathbf{x}}\|^2\}} + \|\hat{\mathbf{x}} - \mathbf{x}_*\| \\ &\stackrel{(b)}{\leq} \hat{d} + \|\hat{\mathbf{x}} - \mathbf{x}_*\| + \sqrt{c^2 + \omega^2}^{nt} \left(\|\mathbf{x}_0 - \hat{\mathbf{x}}\| - \hat{d} \right)^+ \\ &\stackrel{(c)}{\leq} \hat{d} + \|\hat{\mathbf{x}} - \mathbf{x}_*\| \\ &\quad + \sqrt{c^2 + \omega^2}^{nt} \left(\|\mathbf{x}_0 - \mathbf{x}_*\| + \|\hat{\mathbf{x}} - \mathbf{x}_*\| - \hat{d} \right)^+, \end{aligned} \quad (16)$$

where (a) and (c) follow from the Minkowski inequality, and (b) follows from the NC³T.

Equations (15) and (16) show the explicit dependence of $\|\mathbf{x}_t - \mathbf{x}_*\|$ on the initial error $\|\mathbf{x}_0 - \mathbf{x}_*\|$, the linear convergence component, and the error between the fixed point and the desired goal, $\|\hat{\mathbf{x}} - \mathbf{x}_*\|$. This suggests a three-step approach, to be followed when analyzing DGD and diffusion later in the paper: 1) establishing and characterizing the contraction; 2) bounding the error between the fixed point and the desired goal, $\|\hat{\mathbf{x}} - \mathbf{x}_*\|$; and 3) using (15) or (16).

D. Strongly Convex and Smooth Functions

Throughout, we will be working with differentiable, strongly convex, and smooth functions. Several equivalent characterizations of such functions are available [42, Ch. 2].

First-order characterization

A differentiable function $f: \mathbb{R}^m \rightarrow \mathbb{R}$ is μ -strongly convex if

$$\mu\|\mathbf{y} - \mathbf{x}\|^2 \leq (\mathbf{y} - \mathbf{x})^T (\nabla f(\mathbf{y}) - \nabla f(\mathbf{x})) \quad (17)$$

for all $\mathbf{x}$ and $\mathbf{y}$, and L -smooth if

$$\|\nabla f(\mathbf{y}) - \nabla f(\mathbf{x})\| \leq L\|\mathbf{y} - \mathbf{x}\|, \quad (18)$$

for all $\mathbf{x}$ and $\mathbf{y}$.

Note that the application of Cauchy-Schwarz inequality to (17) yields

$$\|\nabla f(\mathbf{y}) - \nabla f(\mathbf{x})\| \geq \mu\|\mathbf{y} - \mathbf{x}\|. \quad (19)$$

Second-order characterization

A *twice* differentiable function $f: \mathbb{R}^m \rightarrow \mathbb{R}$ is μ -strongly convex and L -smooth if

$$L\mathbf{I} \succcurlyeq \nabla^2 f(\mathbf{x}) \succcurlyeq \mu\mathbf{I} \quad (20)$$

for all $\mathbf{x}$. Equation (20) states that $f(\cdot)$ lies between two quadratic functions with curvatures μ and L , respectively.

E. The Mean Hessian Theorem

The gradients of μ -strongly convex, L -smooth functions satisfy the following important relation.

Mean Hessian Theorem (MHT): Let $f: \mathbb{R}^m \rightarrow \mathbb{R}$ be differentiable, μ -strongly convex, and L -smooth. Then for any $\mathbf{x}, \mathbf{y}$, there exists a symmetric matrix $\mathbf{A}_{\mathbf{x}\mathbf{y}}$ (dependent

on x and y such that

$$LI \succcurlyeq A_{xy} \succcurlyeq \mu I, \quad (21)$$

$$\nabla f(y) - \nabla f(x) = A_{xy}(y - x). \quad (22)$$

■

We will prove the MHT in two different ways:

- (i) first, using a line integral argument assuming twice differentiability of $f(\cdot)$; then
- (ii) again, using linear algebra assuming *only (once)* differentiability.

Case (ii) of course subsumes case (i), rendering the proof for case (i) redundant – however, this proof is so intuitive that we give it independently.

Note that in case (ii) the Hessian may not exist, so strictly speaking MHT is a misnomer. Also note that A_{xy} may not be unique. For example, (22) remains satisfied if incrementing or decrementing A_{xy} by the Gramian of any vector that is orthogonal to $y - x$.

1) Proof of MHT for Twice Differentiable $f(\cdot)$

If $f(\cdot)$ is twice differentiable everywhere, the MHT can be established by integrating $\nabla^2 f(\cdot)$ along a line. Let

$$g(\xi) = \nabla f((1 - \xi)x + \xi y).$$

Then

$$g(0) = \nabla f(x),$$

$$g(1) = \nabla f(y),$$

$$g'(\xi) = [\nabla^2 f((1 - \xi)x + \xi y)](y - x),$$

wherefrom

$$\begin{aligned} \nabla f(y) &= g(1) = g(0) + \int_0^1 d\xi g'(\xi) \\ &= \nabla f(x) + \left[\int_0^1 d\xi \nabla^2 f((1 - \xi)x + \xi y) \right] (y - x). \end{aligned} \quad (23)$$

Hence (22) is satisfied with

$$A_{xy} = \int_0^1 d\xi \nabla^2 f((1 - \xi)x + \xi y),$$

which equals the mean Hessian between x and y . Owing to (20),

$$L \geq \frac{z^T [\nabla^2 f((1 - \xi)x + \xi y)] z}{\|z\|^2} \geq \mu, \quad (24)$$

for all ξ and all z , $z \neq 0$. Integration of (24) with respect to ξ over $[0, 1]$ gives

$$L \geq \frac{z^T A_{xy} z}{\|z\|^2} \geq \mu,$$

for all $z \neq 0$, so (21) is satisfied.

2) Proof of MHT for (Only Once) Differentiable $f(x)$

To show the MHT when $f(\cdot)$ is (only) once differentiable everywhere, let

$$a = y - x,$$

$$b = \nabla f(y) - \nabla f(x).$$

If $x = y$ there is nothing to prove, so assume $x \neq y$; then $a \neq 0$ and from (19), $b \neq 0$. From (18) and (19) we have

$$\mu \|a\| \leq \|b\| \leq L \|a\|. \quad (25)$$

Hence,

$$\mu b^T a \stackrel{(a)}{\leq} \mu \|a\| \|b\| \leq \|b\|^2, \quad (26)$$

where (a) follows from the Cauchy-Schwarz inequality.

Equality in (a) in (26) is only possible if a and b are parallel; $a = \|a\|/\|b\| \cdot b$. In this case, take $A_{xy} = \|b\|/\|a\| \cdot I$. Then, because of (25), (21) is satisfied. Since

$$A_{xy} a = A_{xy} \frac{\|a\|}{\|b\|} b = \frac{\|b\|}{\|a\|} \frac{\|a\|}{\|b\|} b = b,$$

(22) is satisfied too.

Now suppose a and b are *not* parallel, so that strict inequality holds in (a) in (26). Then $\|b\|^2 - \mu b^T a > 0$ and we can define

$$\alpha = \frac{b^T a - \mu \|a\|^2}{\|b\|^2 - \mu b^T a}.$$

The use of Theorem 2.1.12 in [42] gives that $b^T a - \mu \|a\|^2 \geq (\|b\|^2 - \mu b^T a)/L$, which implies $\alpha \geq 1/L$. Furthermore, since a and b are not parallel we must have $\|b - \mu a\| > 0$, which implies $\|b\|^2 + \mu^2 \|a\|^2 - 2\mu b^T a > 0$, and in turn that $\alpha < 1/\mu$. Take

$$A_{xy} = \mu \Pi_{(b - \mu a)^\perp} + \frac{1}{\alpha} \Pi_{(b - \mu a)}.$$

Clearly, all eigenvalues of A_{xy} lie between μ and L , so (21) is satisfied. It remains to show that (22) holds:

$$\begin{aligned} A_{xy} a &= \mu \Pi_{(b - \mu a)^\perp} a + \frac{1}{\alpha} \Pi_{(b - \mu a)} a \\ &= \mu a + \left(\frac{1}{\alpha} - \mu \right) \Pi_{(b - \mu a)} a \\ &= \mu a + \left(\frac{1}{\alpha} - \mu \right) \frac{(b - \mu a)^T a}{\|b - \mu a\|^2} (b - \mu a) \\ &= \mu a + \left(\frac{1}{\alpha} - \mu \right) \frac{b^T a - \mu \|a\|^2}{\|b - \mu a\|^2} (b - \mu a) \\ &= \mu a + \left(\frac{1}{\alpha} - \mu \right) \frac{\alpha}{1 - \alpha \mu} (b - \mu a) \\ &= \mu a + b - \mu a = b. \end{aligned} \quad \blacksquare$$

3) Misconceptions

Consider the twice-differentiable case. Equation (23) is reminiscent of the mean value theorem for scalar, differentiable functions [46, Th. 5.10]²; in fact, if x is scalar then A_{xy} is

²If $x \leq z \leq y$, then $g(y) = g(x) + g'(z)(y - x)$.

scalar too and (22) is simply this mean value theorem applied to $f'(\cdot)$. Given this, one may intuit that there exists a ξ such that $\nabla f(\mathbf{y}) = \nabla f(\mathbf{x}) + [\nabla^2 f((1-\xi)\mathbf{x} + \xi\mathbf{y})](\mathbf{y} - \mathbf{x})$. But this is *not* the case, and the assumed existence of such a ξ appears to be a common misconception. For example, this fallacy can be found in papers with analyses of GD algorithms [47, Lemma 5] and, as pointed out in [48], in statistics books and papers.

III. Gradient Descent (GD)

As a preparation for the analysis of DGD and diffusion, we first review the standard GD algorithm for the minimization of a differentiable, μ -strongly convex, and L -smooth function $f: \mathbb{R}^m \rightarrow \mathbb{R}$, and show how the MHT+contraction framework yields a tight convergence bound.

GD minimizes $f(\cdot)$ by computing the sequence of iterates $\{\mathbf{x}_t\}$ according to

$$\mathbf{x}_{t+1} = \mathbf{x}_t - \eta \nabla f(\mathbf{x}_t)$$

for some initial point $\mathbf{x}_0$, and positive step size η . Let

$$\mathbf{x}_* = \underset{\mathbf{x}}{\operatorname{argmin}} f(\mathbf{x})$$

be the optimal point, which is unique because of the strong convexity, and at which $\nabla f(\mathbf{x}_*) = \mathbf{0}$.

To investigate under what conditions $\mathbf{x}_t \rightarrow \mathbf{x}_*$ as $t \rightarrow \infty$, define the mapping

$$\phi(\mathbf{x}) = \mathbf{x} - \eta \nabla f(\mathbf{x}), \quad (27)$$

so that $\mathbf{x}_{t+1} = \phi(\mathbf{x}_t)$. Then for any vectors $\mathbf{x}$ and $\mathbf{y}$,

$$\begin{aligned} & \|\phi(\mathbf{y}) - \phi(\mathbf{x})\| \\ &= \|\mathbf{y} - \eta \nabla f(\mathbf{y}) - [\mathbf{x} - \eta \nabla f(\mathbf{x})]\| \\ &\stackrel{(a)}{=} \|\mathbf{y} - \eta[\nabla f(\mathbf{x}) + \mathbf{A}_{\mathbf{x}\mathbf{y}}(\mathbf{y} - \mathbf{x})] - [\mathbf{x} - \eta \nabla f(\mathbf{x})]\| \\ &= \|(\mathbf{I} - \eta \mathbf{A}_{\mathbf{x}\mathbf{y}})(\mathbf{y} - \mathbf{x})\| \\ &\leq \|\mathbf{I} - \eta \mathbf{A}_{\mathbf{x}\mathbf{y}}\|_2 \|\mathbf{y} - \mathbf{x}\| \\ &\leq \max\{|1 - \eta\mu|, |1 - \eta L|\} \cdot \|\mathbf{y} - \mathbf{x}\| \\ &\stackrel{(b)}{=} \begin{cases} (1 - \eta\mu)\|\mathbf{y} - \mathbf{x}\|, & 0 < \eta \leq \frac{2}{L+\mu}, \\ (\eta L - 1)\|\mathbf{y} - \mathbf{x}\|, & \frac{2}{L+\mu} \leq \eta. \end{cases} \end{aligned} \quad (28)$$

In step (a), we applied the MHT to $f(\cdot)$: for any $\mathbf{x}$ and $\mathbf{y}$ there is a matrix $\mathbf{A}_{\mathbf{x}\mathbf{y}}$ that satisfies (21) and (22). To establish (b), note that $\max\{|1 - \eta\mu|, |1 - \eta L|\} = 1 - \eta\mu$ if and only if $(1 - \eta\mu)^2 \leq (1 - \eta L)^2$, or equivalently,

$$\eta \leq \frac{2}{L + \mu}. \quad (29)$$

Hence, if $0 < \eta \leq 2/(L + \mu)$, $\phi(\cdot)$ is a contraction with factor $1 - \eta\mu < 1$, and if $2/(L + \mu) \leq \eta < 2/L$, $\phi(\cdot)$ is a contraction with factor $\eta L - 1 < 1$. The smallest possible contraction factor equals $(L - \mu)/(L + \mu)$ and is obtained with $\eta = 2/(L + \mu)$. Note that any contraction factor between $(L - \mu)/(L + \mu)$ and 1 can be obtained for two different values of η : one below $2/(L + \mu)$ and one above $2/(L + \mu)$.

Let $\hat{\mathbf{x}}$ be the fixed point of $\phi(\cdot)$. By inspection of (27), $\phi(\mathbf{x}_*) = \mathbf{x}_*$, so the fixed point equals the optimal point: $\hat{\mathbf{x}} = \mathbf{x}_*$. By using the C²T, we conclude that

$$\|\mathbf{x}_t - \mathbf{x}_*\| \leq \begin{cases} (1 - \eta\mu)^t \|\mathbf{x}_0 - \mathbf{x}_*\|, & \eta \leq \frac{2}{L+\mu}, \\ (\eta L - 1)^t \|\mathbf{x}_0 - \mathbf{x}_*\|, & \frac{2}{L+\mu} \leq \eta < \frac{2}{L}. \end{cases} \quad (30)$$

Hence $\mathbf{x}_t \rightarrow \mathbf{x}_*$ as $t \rightarrow \infty$, geometrically fast – that is, GD achieves linear convergence if $\eta < 2/L$.

Tightness of the Bound (30)

To illustrate the tightness of the bound (30), consider GD for $m = 2$, with predetermined η and $\mathbf{x}_0$, applied to $f(\mathbf{x}) = \mu x_1^2/2 + L x_2^2/2$. By direct calculation, $\mathbf{x}_* = \mathbf{0}$, and

$$\mathbf{x}_t = \begin{bmatrix} 1 - \eta\mu & 0 \\ 0 & 1 - \eta L \end{bmatrix}^t \mathbf{x}_0.$$

If $\eta \geq 2/L$, then with $\mathbf{x}_0 = [0, 1]^T$, $\mathbf{x}_t$ oscillates between $[0, 1]^T$ and $[0, -1]^T$, or diverges. If $2/(L + \mu) \leq \eta < 2/L$, the worst-case choice of $\mathbf{x}_0$ is $\mathbf{x}_0 = [0, 1]^T$, which gives $\|\mathbf{x}_t - \mathbf{x}_*\| = (\eta L - 1)^t \|\mathbf{x}_0 - \mathbf{x}_*\|$. Finally, if $\eta \leq 2/(L + \mu)$, the worst-case $\mathbf{x}_0$ is $\mathbf{x}_0 = [1, 0]^T$, and $\|\mathbf{x}_t - \mathbf{x}_*\| = (1 - \eta\mu)^t \|\mathbf{x}_0 - \mathbf{x}_*\|$. Taken together, this shows that for a given step size $\eta < 2/L$, there exist a worst-case function $f(\cdot)$ and initialization $\mathbf{x}_0$ such that the bound in (30) is met with equality. Conversely, when $\eta \geq 2/L$, GD either does not converge or diverges, in the worst case. We conclude that (30) characterizes the GD contraction over the entire range of η for which convergence is guaranteed.

We shall be mainly interested in operating regimes where η can be arbitrarily small. For GD, the relevant step size condition is then (29), under which (30) gives

$$\|\mathbf{x}_t - \mathbf{x}_*\| \leq (1 - \eta\mu)^t \|\mathbf{x}_0 - \mathbf{x}_*\|. \quad (31)$$

Note that the contraction factor in (31), $1 - \eta\mu$, is only dependent on μ but not on L ; however, the condition on η in (29) depends on L . To the authors' knowledge, existing derivations of the bound in (31) with contraction factor $1 - \eta\mu$ rely on *first-order characterizations* of strong convexity and smoothness; see, for example [49, Th. 3.12]. In contrast, our MHT+contraction approach uses a description akin to *second-order characterizations* of twice differentiable functions, but without requiring twice differentiability. The powerfulness of this approach is that when bounding the eigenvalues of $\mathbf{I} - \eta \mathbf{A}_{\mathbf{x}\mathbf{y}}$, the strong convexity and smoothness properties of $f(\cdot)$ are used simultaneously, which immediately gives (28). Some analyses in the literature that use these properties separately yield looser bounds; see, for example, [50, Th. 2 (Ch. 1)], [51, Th. 3.10], [21, Lemma 1], and [30, Th. 1].

Unboundedness of $\nabla f(\mathbf{x})$

A common misconception is that the gradients of strongly convex functions can be uniformly bounded over $\mathbb{R}^m$. For example, this assumption is made in [52, Th. 14.11], and in [27], [30], [31], [53]–[56] (for distributed GD ap-

plied to federated learning). But as pointed out for example in [57], a strongly convex function always has unbounded gradients over $\mathbb{R}^m$. This is a simple consequence of (19): if $f(\cdot)$ is μ -strongly convex then $\|\nabla f(\mathbf{x})\| = \|\nabla f(\mathbf{x}) - \nabla f(\mathbf{x}_*)\| \geq \mu\|\mathbf{x} - \mathbf{x}_*\|$, which is unbounded on $\mathbb{R}^m$.

Importantly, our MHT+contraction analysis framework requires no assumptions on boundedness of the gradients.

IV. Decentralized Gradient Descent (DGD) and Diffusion

This section contains the main analysis of DGD and diffusion. We first describe the algorithms, and then analyze their convergence properties using the MHT, C²T, and C³T, both for single and multiple local gradient updates per iteration.

A. Communication Network Topology

We consider N agents interconnected in a communication network defined by its adjacency (weight) matrix $\mathbf{W}$. We assume that this network is connected, undirected ($\mathbf{W}^T = \mathbf{W}$), and that the weights sum up to one at each agent so that $\mathbf{W}\mathbf{1} = \mathbf{1}$. We also assume that the (real-valued) eigenvalues of $\mathbf{W}$, $\{\lambda_n\}$, when ordered in descending order, satisfy

$$1 = \lambda_1 > \lambda_2 \geq \dots \geq \lambda_N > -1. \quad (32)$$

Note that $\mathbf{W}$ is *not* required to be non-negative, and not required to be positive (semi-)definite. The case of $\lambda_N = -1$ is discussed in Sections VII.E–D.

Under the assumptions made, we have that $\mathbf{W}^t \rightarrow \mathbf{1}\mathbf{1}^T/N$ as $t \rightarrow \infty$, which implies $\mathbf{W}^t \mathbf{x} \rightarrow \mathbf{1}\mathbf{1}^T \mathbf{x}/N$ for any $\mathbf{x}$. For this reason, the multiplication of a vector $\mathbf{x}$ with $\mathbf{W}$ is known as *consensus averaging*.

B. The DGD and Diffusion Algorithms

All relevant structural aspects of the problem surface for a scalar parameter, considered here. Let $f_n : \mathbb{R} \rightarrow \mathbb{R}$ be the local objective function of the n th agent; herein, we simply treat each $f_n(\cdot)$ as a given function while in an application, it would be dependent on data available at the n th agent. We assume that all $\{f_n(\cdot)\}$ are μ -strongly convex and L -smooth. The goal is to minimize the global objective function

$$F(x) = \frac{1}{N} \sum_{n=1}^N f_n(x),$$

which has the unique minimum

$$x_* = \underset{x}{\operatorname{argmin}} F(x),$$

at which $F'(x_*) = 0$.

For future use, we introduce the following “component-wise” objective function

$$f(\mathbf{x}) = \sum_{n=1}^N f_n(x_n),$$

where $\mathbf{x} = [x_1, \dots, x_N]^T$, and define

$$\mathbf{x}_* = x_* \mathbf{1} = [x_*, \dots, x_*]^T.$$

(This vector $\mathbf{x}_*$ must not be conflated with x_* in Section III; henceforth we write $\mathbf{x}_*$ in lieu of $x_* \mathbf{1}$ for brevity.) Note that

$$\mathbf{W}\mathbf{x}_* = \mathbf{x}_*, \quad (33)$$

$$\nabla f(\mathbf{x}) = [f'_1(x_1), \dots, f'_N(x_N)]^T,$$

$$\mathbf{1}^T \nabla f(\mathbf{x}_*) = \sum_{n=1}^N f'_n(x_*) = NF'(\mathbf{x}_*) = 0, \quad (34)$$

but that $\nabla f(\mathbf{x}_*) \neq \mathbf{0}$ in general.

DGD and diffusion work by having each agent n forming a sequence of iterates. We let $\mathbf{x}_t$ be the N -vector whose n th component represents the iterate of the n th agent at time t . The iterates are formed by consensus averaging among the agents’ neighbors using the weights in $\mathbf{W}$, and subtraction of a local gradient update. There are three basic versions of the algorithms:

- *Standard DGD* (called “consensus strategy” in [43]):

$$\mathbf{x}_{t+1} = \mathbf{W}\mathbf{x}_t - \eta \nabla f(\mathbf{x}_t), \quad (35)$$

where η is a positive step size. Here each agent computes its local gradient update based on the iterate available at the beginning of the iteration. This allows agents to compute their gradient updates while simultaneously communicating with their neighbors.

- *Adapt-then-combine (ATC) diffusion* (e.g., [8], [58]):

$$\mathbf{x}_{t+1} = \mathbf{W}(\mathbf{x}_t - \eta \nabla f(\mathbf{x}_t)). \quad (36)$$

Here too, each agent computes its local gradient update based on the iterate available at the beginning of the iteration, but in contrast to the standard DGD, these updates are weighted by $\mathbf{W}$.

- *Combine-then-adapt (CTA) diffusion* (e.g., [58]),

$$\mathbf{x}_{t+1} = \mathbf{W}\mathbf{x}_t - \eta \nabla f(\mathbf{W}\mathbf{x}_t).$$

Here, each agent computes its gradient update *after* forming consensus with the incoming iterates from the neighbors.

ATC and CTA diffusion are closely interrelated. In fact, CTA diffusion becomes equivalent to ATC diffusion if $\mathbf{x}_t$ is re-defined to be the output after applying $\mathbf{W}$. Therefore, we will henceforth only consider the standard DGD in (35), and ATC diffusion in (36), and use the names *DGD* and *diffusion* for these two algorithms. Defining the matrix

$$\mathbf{Z} = \begin{cases} \mathbf{I}, & \text{for DGD} \\ \mathbf{W}, & \text{for diffusion,} \end{cases} \quad (37)$$

we can write the DGD and diffusion iterations (35) and (36) as

$$\mathbf{x}_{t+1} = \mathbf{W}\mathbf{x}_t - \eta \mathbf{Z} \nabla f(\mathbf{x}_t). \quad (38)$$

C. Convergence to a Fixed Point

We first show that for both DGD and diffusion the iterate $\mathbf{x}_t$ converges geometrically to a fixed point (to be denoted

$\hat{x}$). Towards this end, define the mapping

$$\phi(\mathbf{x}) = \mathbf{W}\mathbf{x} - \eta \mathbf{Z} \nabla f(\mathbf{x}),$$

so that (38) can be written

$$\mathbf{x}_{t+1} = \phi(\mathbf{x}_t). \quad (39)$$

1) [Standard] DGD

Consider DGD ($\mathbf{Z} = \mathbf{I}$). For any vectors $\mathbf{x}$ and $\mathbf{y}$,

$$\begin{aligned} & \|\phi(\mathbf{y}) - \phi(\mathbf{x})\| \\ &= \|\mathbf{W}\mathbf{y} - \eta \nabla f(\mathbf{y}) - [\mathbf{W}\mathbf{x} - \eta \nabla f(\mathbf{x})]\| \\ &\stackrel{(a)}{=} \|\mathbf{W}\mathbf{y} - \eta[\nabla f(\mathbf{x}) + \mathbf{A}_{\mathbf{x}\mathbf{y}}(\mathbf{y} - \mathbf{x})] - [\mathbf{W}\mathbf{x} - \eta \nabla f(\mathbf{x})]\| \\ &= \|(\mathbf{W} - \eta \mathbf{A}_{\mathbf{x}\mathbf{y}})(\mathbf{y} - \mathbf{x})\| \\ &\stackrel{(b)}{\leq} \max\{|\lambda_N - \eta L|, |1 - \eta\mu|\} \cdot \|\mathbf{y} - \mathbf{x}\| \\ &= \begin{cases} (1 - \eta\mu)\|\mathbf{y} - \mathbf{x}\|, & 0 < \eta \leq \frac{1+\lambda_N}{L+\mu}, \\ (\eta L - \lambda_N)\|\mathbf{y} - \mathbf{x}\|, & \frac{1+\lambda_N}{L+\mu} \leq \eta, \end{cases} \end{aligned} \quad (40)$$

where in (a) we applied the MHT to $f(\cdot)$ and in (b) we used

$$(\lambda_N - \eta L)\mathbf{I} \preceq \mathbf{W} - \eta \mathbf{A}_{\mathbf{x}\mathbf{y}} \preceq (1 - \eta\mu)\mathbf{I}.$$

Hence $\phi(\cdot)$ is a contraction with factor $1 - \eta\mu < 1$ if

$$\eta \leq \frac{1 + \lambda_N}{L + \mu}, \quad (41)$$

and a contraction with factor $\eta L - \lambda_N < 1$ if $(1 + \lambda_N)/(L + \mu) \leq \eta < (1 + \lambda_N)/L$.

2) Diffusion

For diffusion, $\mathbf{Z} = \mathbf{W}$. Then for any $\mathbf{x}$ and $\mathbf{y}$,

$$\begin{aligned} \|\phi(\mathbf{y}) - \phi(\mathbf{x})\| &= \|\mathbf{W}(\mathbf{y} - \eta \nabla f(\mathbf{y}) - [\mathbf{x} - \eta \nabla f(\mathbf{x})])\| \\ &\leq \|\mathbf{W}\|_2 \|\mathbf{y} - \eta \nabla f(\mathbf{y}) - [\mathbf{x} - \eta \nabla f(\mathbf{x})]\| \\ &\stackrel{(a)}{=} \|(\mathbf{I} - \eta \mathbf{A}_{\mathbf{x}\mathbf{y}})(\mathbf{y} - \mathbf{x})\| \\ &\stackrel{(b)}{\leq} \begin{cases} (1 - \eta\mu)\|\mathbf{y} - \mathbf{x}\|, & 0 < \eta \leq \frac{2}{L+\mu}, \\ (\eta L - 1)\|\mathbf{y} - \mathbf{x}\|, & \frac{2}{L+\mu} \leq \eta, \end{cases} \end{aligned} \quad (42)$$

where in (a) we used that $\|\mathbf{W}\|_2 = 1$ and in (b) we used the calculation in Section III. The conditions on η for $\phi(\cdot)$ to be a contraction are the same as for GD.

3) Convergence to a Fixed Point

For both DGD and diffusion, the C²T shows that under the stated conditions on η , the iteration (38) converges to fixed point, $\hat{x}$ such that

$$\hat{x} = \mathbf{W}\hat{x} - \eta \mathbf{Z} \nabla f(\hat{x}). \quad (43)$$

Specifically, for DGD:

$$\|\mathbf{x}_t - \hat{x}\| \leq \begin{cases} (1 - \eta\mu)^t \|\mathbf{x}_0 - \hat{x}\|, & 0 < \eta \leq \frac{1+\lambda_N}{L+\mu}, \\ (\eta L - \lambda_N)^t \|\mathbf{x}_0 - \hat{x}\|, & \frac{1+\lambda_N}{L+\mu} \leq \eta < \frac{1+\lambda_N}{L}, \end{cases} \quad (44)$$

and for diffusion:

$$\|\mathbf{x}_t - \hat{x}\| \leq \begin{cases} (1 - \eta\mu)^t \|\mathbf{x}_0 - \hat{x}\|, & 0 < \eta \leq \frac{2}{L+\mu}, \\ (\eta L - 1)^t \|\mathbf{x}_0 - \hat{x}\|, & \frac{2}{L+\mu} \leq \eta < \frac{2}{L}. \end{cases} \quad (45)$$

4) Tightness of the Bounds (44) and (45)

To appreciate the tightness of these bounds, consider DGD applied to $f_n(x) = \rho x^2/2$, for some $\rho > 0$. Then $\hat{x} = \mathbf{0}$ and

$$\mathbf{x}_t - \hat{x} = (\mathbf{W} - \eta\rho\mathbf{I})^t(\mathbf{x}_0 - \hat{x}).$$

Let $\mathbf{u}_n$ be a unit-norm eigenvector of $\mathbf{W}$ associated with the eigenvalue λ_n , and note that with the initial point $\mathbf{x}_0 = \mathbf{u}_n$,

$$\mathbf{x}_t - \hat{x} = (\mathbf{W} - \eta\rho\mathbf{I})^t \mathbf{u}_n = (\lambda_n - \eta\rho)^t \mathbf{u}_n.$$

If $\eta \geq (1 + \lambda_N)/L$, $\rho = L$, and $\mathbf{x}_0 = \mathbf{u}_N$, $\mathbf{x}_t$ either oscillates between $\mathbf{u}_N$ and $-\mathbf{u}_N$, or diverges. If $(1 + \lambda_N)/(L + \mu) \leq \eta < (1 + \lambda_N)/L$, $\rho = L$, and $\mathbf{x}_0 = \mathbf{u}_N$, we have $\|\mathbf{x}_t - \hat{x}\| = (\eta L - \lambda_N)^t$. Finally, if $\eta \leq (1 + \lambda_N)/(L + \mu)$, $\rho = \mu$, and $\mathbf{x}_0 = \mathbf{u}_1 = \mathbf{1}/\sqrt{N}$, we have $\|\mathbf{x}_t - \hat{x}\| = (1 - \eta\mu)^t$. This shows that for arbitrary $\mathbf{W}$, and a given step size $\eta < (1 + \lambda_N)/L$, there is a set of worst-case functions $\{f_n(\cdot)\}$ and an initial point $\mathbf{x}_0$ for which (44) is met with equality. Conversely, when $\eta \geq (1 + \lambda_N)/L$, DGD may either not converge, or diverge.

A similar example can be constructed to demonstrate the tightness of the bound for diffusion, (45).

We conclude that (44) and (45) characterize the DGD and diffusion contractions over the entire range of η for which convergence to a fixed point is guaranteed.

D. Distance from the Fixed Point to the Optimum

We next upper-bound the distance from the fixed point to the optimum, $\|\hat{x} - \mathbf{x}_*\|$. To this end, note from (43) that

$$\begin{aligned} \mathbf{0} &= \frac{1}{\eta}(\mathbf{I} - \mathbf{W})\hat{x} + \mathbf{Z} \nabla f(\hat{x}) \\ &\stackrel{(a)}{=} \frac{1}{\eta}(\mathbf{I} - \mathbf{W})\hat{x} + \mathbf{Z} \nabla f(\mathbf{x}_*) + \mathbf{Z} \mathbf{A}_{\mathbf{x}_* \hat{x}}(\hat{x} - \mathbf{x}_*) \\ &\stackrel{(b)}{=} \frac{1}{\eta}(\mathbf{I} - \mathbf{W})(\hat{x} - \mathbf{x}_*) + \mathbf{Z} \nabla f(\mathbf{x}_*) + \mathbf{Z} \mathbf{A}_{\mathbf{x}_* \hat{x}}(\hat{x} - \mathbf{x}_*) \\ &= \left[\mathbf{Z} \mathbf{A}_{\mathbf{x}_* \hat{x}} + \frac{1}{\eta}(\mathbf{I} - \mathbf{W}) \right] (\hat{x} - \mathbf{x}_*) + \mathbf{Z} \nabla f(\mathbf{x}_*), \end{aligned} \quad (46)$$

where $\mathbf{A}_{\mathbf{x}_* \hat{x}}$ satisfies (21); in (a) we applied the MHT to $f(\cdot)$, and in (b) we used (33). Now apply the distance-bounding result in Appendix A to (46), with $\mathbf{d} = \hat{x} - \mathbf{x}_*$ and $\mathbf{v} = \nabla f(\mathbf{x}_*)$, and use (34). This yields

$$\|\hat{x} - \mathbf{x}_*\| \leq \eta \frac{L}{\mu} \Lambda \|\nabla f(\mathbf{x}_*)\|, \quad (47)$$

where we define the “topology factor”

$$\Lambda = \begin{cases} \|(I - W)^\dagger\|_2 = \frac{1}{1 - \lambda_2}, & \text{for DGD} \\ 2\|(I - W)^\dagger W\|_2, & \text{for diffusion,} \end{cases} \quad (48)$$

and where for diffusion we additionally require $\eta \leq 1/(L\Lambda)$.

The right hand side of (47) consists of four factors: η , which must be small for $\hat{x}$ to be close to x_* ; the condition number L/μ , which depends on the convexity and smoothness properties of the local objectives; Λ , which differs between DGD and diffusion; and $\|\nabla f(x_*)\|$ that captures the data heterogeneity.

E. Bounding the Total Error

Henceforth, we will be interested in criteria that guarantee convergence for arbitrarily small η and hence impose (41) for DGD and (29) for diffusion, leading to

$$\|x_t - \hat{x}\| \leq (1 - \eta\mu)^t \|x_0 - \hat{x}\| \quad (49)$$

for both algorithms. Readers interested in the “upper step size regimes” $[(1 + \lambda_N)/L > \eta \geq (1 + \lambda_N)/(L + \mu)$ for DGD, and $2/L > \eta \geq 2/(L + \mu)$ for diffusion, respectively] may simply replace $1 - \eta\mu$ by $\eta L - \lambda_N$ or $\eta L - 1$ wherever pertinent in the subsequent development.

From (15) and (49), we obtain a bound on the total error:

$$\|x_t - x_*\| \leq (1 - \eta\mu)^t \|x_0 - \hat{x}\| + \|\hat{x} - x_*\|. \quad (50)$$

For any given $\epsilon > 0$, there exists η such that the second term of (50) is less than $\epsilon/2$. For that η , there is a value of t such that the first term is less than $\epsilon/2$; thus $\|x_t - x_*\| \leq \epsilon$. In particular, η should be of order $\sim \epsilon$ to make the second term in (50) small. Since $1 - \eta\mu \leq e^{-\eta\mu}$, t should be of order $\sim \log(1/\epsilon)/\epsilon$ to make the geometric factor $(1 - \eta\mu)^t$ small.

F. Comparison Between DGD and Diffusion

The convergence properties of DGD and diffusion have a complicated dependence on the curvature of $\{f_n(\cdot)\}$ and on the network topology. Some intuition for the dependence on topology can be gained by considering the distance $\|\hat{x} - x_*\|$ in (47). This distance scales with $1/(1 - \lambda_2)$ for DGD and with $\|(I - W)^\dagger W\|_2$ for diffusion. Since

$$\|(I - W)^\dagger W\|_2 \leq \|(I - W)^\dagger\|_2 = \frac{1}{1 - \lambda_2}, \quad (51)$$

diffusion has a more favorable dependence on the topology than DGD.

While DGD operates under the step size condition (41), diffusion requires the condition (29). Additionally, for diffusion, (47) requires $\eta \leq 1/(L\Lambda)$. These conditions can differ significantly if λ_2 is close to 1 or λ_N is close to -1 .

With DGD, the communication between agents can take place in parallel with the gradient computations, whereas this is not possible for diffusion. However, this difference between the two algorithms is only significant if the time spent on communication and the total time spent on compu-

tation (counting all local gradient updates, see Section IV.G) are of comparable order.

The Special Case of a Complete Graph

For the special case of a complete graph, $W = \mathbf{1}\mathbf{1}^T/N$. Then $\lambda_2 = \dots = \lambda_N = 0$, and $(I - W)^\dagger W = 0$.

- For DGD, (47) specializes to

$$\|\hat{x} - x_*\| \leq \eta \frac{L}{\mu} \|\nabla f(x_*)\|.$$

- For diffusion, $\Lambda = 0$ so (47) holds with equality. In this case the fixed point equals the optimal point: $\hat{x} = x_*$. This can also be seen by setting $Z = W$ in (43):

$$x_* \stackrel{(a)}{=} W x_* \stackrel{(b)}{=} W x_* - \eta W \nabla f(x_*),$$

where in (a) we used (33) and in (b) we used (34).

- The step size condition for DGD, (41), becomes $\eta \leq 1/(L + \mu)$ – to be compared with $\eta \leq 2/(L + \mu)$ for diffusion. (The condition $\eta \leq 1/(L\Lambda)$ vanishes.)

G. Multiple Local Gradient Updates

The agents may perform multiple, say T , local gradient updates before averaging the iterates with their neighbors. We next extend the analyses to this case.

Let $x_{t,i}$ be a vector whose n th component denotes the iterate at agent n at time t , after the i th local gradient update. Using these auxiliary variables, we can describe DGD and diffusion with T local gradient updates as follows:

$$x_{t,0} = x_t, \quad (52)$$

$$x_{t,1} = x_{t,0} - \eta \nabla f(x_{t,0}),$$

$$x_{t,2} = x_{t,1} - \eta \nabla f(x_{t,1}),$$

...

$$x_{t,T-1} = x_{t,T-2} - \eta \nabla f(x_{t,T-2}),$$

$$x_{t+1} = W x_{t,T-1} - \eta Z \nabla f(x_{t,T-1}). \quad (53)$$

Note that DGD and diffusion only differ in the last step, where the final local gradient update is performed.

1) Convergence to a Fixed Point $\hat{x}$

We can write (52)–(53) as

$$x_{t+1} = \varphi(x_t), \quad (54)$$

where $\varphi(x) = \phi_T(\dots(\phi_1(x)))$, and the mappings $\{\phi_1(\cdot), \dots, \phi_T(\cdot)\}$ are defined through

$$\phi_i(x) = x - \eta \nabla f(x), \quad i = 1, \dots, T - 1 \quad (55)$$

$$\phi_T(x) = W x - \eta Z \nabla f(x). \quad (56)$$

Under suitable step size conditions, $\varphi(\cdot)$ in (54) is a contraction. To show this, note that by the calculation in Section III, $\{\phi_1(\cdot), \dots, \phi_{T-1}(\cdot)\}$ are contractions, with factor $1 - \eta\mu$. Next, for DGD, a calculation identical to (40) shows that if (41) holds, $\phi_T(\cdot)$ is a contraction with

Algorithm	DGD, $T = 1$	diffusion, $T = 1$	DGD, $T > 1$	diffusion, $T > 1$
$\ x_t - \hat{x}\ $	(49)	(49)	(57)	(57)
$\ \hat{x} - x_*\ $	(47) and (48)	(47) and (48)	(58)	(59)
requirement on η	(41)	(29) and $\eta \leq 1/(L\Lambda)$	(41)	(29)

TABLE 1. Summary of the main convergence bounds for DGD and diffusion.

factor $1 - \eta\mu$. For diffusion, the calculation in (42) shows that if (29) holds, $\phi_T(\cdot)$ is a contraction with factor $1 - \eta\mu$. The C^3T then shows that $\varphi(\cdot)$ is a contraction, with factor $(1 - \eta\mu)^T$. Hence, x_t converges to a fixed point, $\hat{x}$, which satisfies $\hat{x} = \varphi(\hat{x})$. Also,

$$\|x_t - \hat{x}\| \leq (1 - \eta\mu)^{Tt} \|x_0 - \hat{x}\|. \quad (57)$$

2) Distance from $\hat{x}$ to x_*

The bounding of $\|\hat{x} - x_*\|$ is complicated by the nonlinear nature of the nested mappings in (52)–(53). In Appendix B we obtain the bounds (58) for DGD, and (59) for diffusion, shown on top of the next page.

Note from (58)–(59) that the bound on $\|\hat{x} - x_*\|$ increases linearly with increasing T . With additional local gradient updates, the iterates at different agents diverge from each other and move towards the respective minimizers of the local objectives $\{f_n(\cdot)\}$. The only remedy is to reduce η .

3) The Special Case of a Federated Learning Setup

In federated learning, after every iteration each agent transmits its iterate to a central server, whereafter the server averages these iterates and broadcasts the average back to the agents. Typically, each agent performs $T > 1$ local gradient updates in each iteration. This results in the algorithm

$$x_{t,0} = x_t, \quad (60)$$

$$x_{t,i+1} = x_{t,i} - \eta \nabla f(x_{t,i}), \quad i = 0, \dots, T-1,$$

$$x_t = \frac{1}{N} \mathbf{1}^T \cdot x_{t,T}. \quad (61)$$

Equations (60)–(61) are equivalent to (52)–(53) with $Z = W = \mathbf{1}\mathbf{1}^T/N$ (and using an auxiliary variable $x_{t,T}$). Thus federated learning is an instance of diffusion on a complete-graph topology, with multiple local gradient updates. Hence, (59) applies. Since $(I - W)^\dagger W = 0$, (59) reduces to

$$\|\hat{x} - x_*\| \leq \eta \frac{T-1}{2} \frac{L}{\mu} \|\nabla f(x_*)\| + \mathcal{O}(\eta^2 T^2).$$

4) The Special Case of $T = 1$

In the special case of a single local gradient update ($T = 1$), (54) reduces to (39), and the bounds (58) and (59) reduce to

$$\|\hat{x} - x_*\| \leq \eta \left(1 + \frac{L}{\mu}\right) \frac{1}{1 - \lambda_2} \|\nabla f(x_*)\| + \mathcal{O}(\eta^2) \quad (62)$$

for DGD, and

$$\|\hat{x} - x_*\| \leq \eta \left(1 + \frac{L}{\mu}\right) \|(I - W)^\dagger W\|_2 \|\nabla f(x_*)\| + \mathcal{O}(\eta^2) \quad (63)$$

for diffusion. Equations (62)–(63) may be compared to (47), which was obtained using a different approach.

H. Summary and Comments

Table 1 summarizes the bounds obtained so far.

Many alternative analysis approaches exist. For example, as shown in [7], the DGD iteration (35) is equivalent to GD on the Lyapunov function $f(x) + x^T(I - W)x/(2\eta)$. However, for the diffusion iteration (36), no such Lyapunov equivalent exists to our knowledge.

For DGD, comparing our bound on $\|x_t - x_*\|$ with [7, Cor. 9], one can show that if η satisfies (41), our contraction speed is faster than the linear convergence factor in [7]; that is, $c_3 \geq 1 - \eta\mu$, with c_3 as in [7]. One can also show that for values of η permitted by [7, Cor. 9], the bound on the asymptotic error therein (the second plus third term in [7, Cor. 9]) is larger than our bound on $\|\hat{x} - x_*\|$ in (47).

I. Time-Varying Step Size

The analyses extend directly to the case when the step size depends on the iteration index. We limit the discussion here to the case of a single local gradient update.

Let η_t be the step size used in the t th iteration, and let $\hat{x}_t$ be the fixed point associated with step size η_t ; this is the $\hat{x}$ that satisfies (43) with η replaced by η_t . The iteration (38) now “chases a moving target” (a moving fixed point). Note that the fixed point changes from one iteration to the next, but it is a fixed point in the sense of the limit of x_t if (38) were iterated with the same step size.

We will need an upper bound on how much the fixed point moves between two consecutive iterations, $\|\hat{x}_{t+1} - \hat{x}_t\|$. To this end, note from (43) that

$$0 = Z \nabla f(\hat{x}_{t+1}) + \frac{1}{\eta_{t+1}} (I - W) \hat{x}_{t+1}, \quad (64)$$

$$\text{and} \quad 0 = Z \nabla f(\hat{x}_t) + \frac{1}{\eta_t} (I - W) \hat{x}_t. \quad (65)$$

$$\text{DGD: } \|\hat{\mathbf{x}} - \mathbf{x}_*\| \leq \eta \left[\frac{T-1}{2} \frac{L}{\mu} + \left(1 + \frac{L}{\mu}\right) \left((T-1) \|(I - \mathbf{W})^\dagger \mathbf{W}\|_2 + \frac{1}{1 - \lambda_2} \right) \right] \|\nabla f(\mathbf{x}_*)\| + \mathcal{O}(\eta^2 T^2) \quad (58)$$

$$\text{diffusion: } \|\hat{\mathbf{x}} - \mathbf{x}_*\| \leq \eta \left[\frac{T-1}{2} \frac{L}{\mu} + T \left(1 + \frac{L}{\mu}\right) \|(I - \mathbf{W})^\dagger \mathbf{W}\|_2 \right] \|\nabla f(\mathbf{x}_*)\| + \mathcal{O}(\eta^2 T^2) \quad (59)$$

Applying the MHT to $f(\cdot)$ in (65) gives

$$\begin{aligned} \mathbf{0} &= \mathbf{Z} \nabla f(\hat{\mathbf{x}}_{t+1}) + \mathbf{Z} \mathbf{A}_{\hat{\mathbf{x}}_{t+1} \hat{\mathbf{x}}_t} (\hat{\mathbf{x}}_t - \hat{\mathbf{x}}_{t+1}) + \frac{1}{\eta_t} (\mathbf{I} - \mathbf{W}) \hat{\mathbf{x}}_t \\ &\stackrel{(a)}{=} \frac{-1}{\eta_{t+1}} (\mathbf{I} - \mathbf{W}) \hat{\mathbf{x}}_{t+1} + \mathbf{Z} \mathbf{A}_{\hat{\mathbf{x}}_{t+1} \hat{\mathbf{x}}_t} (\hat{\mathbf{x}}_t - \hat{\mathbf{x}}_{t+1}) \\ &\quad + \frac{1}{\eta_t} (\mathbf{I} - \mathbf{W}) \hat{\mathbf{x}}_t \\ &= \left[\mathbf{Z} \mathbf{A}_{\hat{\mathbf{x}}_{t+1} \hat{\mathbf{x}}_t} + \frac{1}{\eta_{t+1}} (\mathbf{I} - \mathbf{W}) \right] (\hat{\mathbf{x}}_t - \hat{\mathbf{x}}_{t+1}) \\ &\quad + \left(\frac{1}{\eta_t} - \frac{1}{\eta_{t+1}} \right) (\mathbf{I} - \mathbf{W}) \hat{\mathbf{x}}_t \\ &\stackrel{(b)}{=} \left[\mathbf{Z} \mathbf{A}_{\hat{\mathbf{x}}_{t+1} \hat{\mathbf{x}}_t} + \frac{1}{\eta_{t+1}} (\mathbf{I} - \mathbf{W}) \right] (\hat{\mathbf{x}}_t - \hat{\mathbf{x}}_{t+1}) \\ &\quad - \eta_t \left(\frac{1}{\eta_t} - \frac{1}{\eta_{t+1}} \right) \mathbf{Z} \nabla f(\hat{\mathbf{x}}_t), \end{aligned}$$

where in (a) we used (64), and in (b) we used again (65).

Now apply the distance bounding result in Appendix A with $\mathbf{d} = \hat{\mathbf{x}}_t - \hat{\mathbf{x}}_{t+1}$ and

$$\mathbf{v} = -\eta_t \left(\frac{1}{\eta_t} - \frac{1}{\eta_{t+1}} \right) \nabla f(\hat{\mathbf{x}}_t),$$

noting that $\Pi_1 \nabla f(\hat{\mathbf{x}}_t) = \mathbf{0}$. We obtain, under appropriate conditions on η_t :

$$\begin{aligned} \|\hat{\mathbf{x}}_{t+1} - \hat{\mathbf{x}}_t\| &\leq \left| \frac{1}{\eta_t} - \frac{1}{\eta_{t+1}} \right| \eta_{t+1} \eta_t \frac{L}{\mu} \Lambda \|\nabla f(\hat{\mathbf{x}}_t)\| \\ &\stackrel{(a)}{\leq} \left| \frac{1}{\eta_t} - \frac{1}{\eta_{t+1}} \right| \eta_{t+1} \eta_t \frac{L}{\mu} \Lambda \cdot 2 \|\nabla f(\mathbf{x}_*)\| \\ &= 2 \left| \eta_t - \eta_{t+1} \right| \frac{L}{\mu} \Lambda \|\nabla f(\mathbf{x}_*)\|, \end{aligned} \quad (66)$$

where in (a) we used that

$$\begin{aligned} \|\nabla f(\hat{\mathbf{x}}_t)\| &= \|\nabla f(\mathbf{x}_*) + \mathbf{A}_{\mathbf{x}_* \hat{\mathbf{x}}_t} (\hat{\mathbf{x}}_t - \mathbf{x}_*)\| \\ &\leq \|\nabla f(\mathbf{x}_*)\| + \|\mathbf{A}_{\mathbf{x}_* \hat{\mathbf{x}}_t} (\hat{\mathbf{x}}_t - \mathbf{x}_*)\| \\ &\leq \|\nabla f(\mathbf{x}_*)\| + L \|\hat{\mathbf{x}}_t - \mathbf{x}_*\| \\ &\stackrel{(a)}{\leq} \|\nabla f(\mathbf{x}_*)\| + \eta_t L \frac{L}{\mu} \Lambda \|\nabla f(\mathbf{x}_*)\| \\ &\leq 2 \|\nabla f(\mathbf{x}_*)\|. \end{aligned} \quad (67)$$

In (67), (a) is a consequence of (47) for small enough η_t .

With a bound on the fixed point drift over time, we bound the distance between the iterates and the fixed points as

$$\begin{aligned} \|\mathbf{x}_{t+1} - \hat{\mathbf{x}}_{t+1}\| &\stackrel{(a)}{\leq} \|\mathbf{x}_{t+1} - \hat{\mathbf{x}}_t\| + \|\hat{\mathbf{x}}_t - \hat{\mathbf{x}}_{t+1}\| \\ &\leq (1 - \eta_t \mu) \|\mathbf{x}_t - \hat{\mathbf{x}}_t\| + \|\hat{\mathbf{x}}_t - \hat{\mathbf{x}}_{t+1}\| \\ &\stackrel{(b)}{\leq} (1 - \eta_t \mu) \|\mathbf{x}_t - \hat{\mathbf{x}}_t\| + 2 \left| \eta_t - \eta_{t+1} \right| \frac{L}{\mu} \Lambda \|\nabla f(\mathbf{x}_*)\|, \end{aligned}$$

where (a) follows from the triangle inequality, and (b) from (66). By induction over t , and using that $\|\mathbf{x}_0 - \hat{\mathbf{x}}_0\| \leq \|\mathbf{x}_0 - \mathbf{x}_*\| + \|\hat{\mathbf{x}}_0 - \mathbf{x}_*\|$, we obtain (68), on the next page, where $\|\hat{\mathbf{x}}_0 - \mathbf{x}_*\|$ is bounded by (47). Finally, we bound the total error as

$$\begin{aligned} \|\mathbf{x}_t - \mathbf{x}_*\| &\stackrel{(a)}{\leq} \|\mathbf{x}_t - \hat{\mathbf{x}}_t\| + \|\hat{\mathbf{x}}_t - \mathbf{x}_*\| \\ &\stackrel{(b)}{\leq} \|\mathbf{x}_t - \hat{\mathbf{x}}_t\| + \eta_t \frac{L}{\mu} \Lambda \|\nabla f(\mathbf{x}_*)\|, \end{aligned}$$

where (a) is the triangle inequality, (b) follows from (47), and $\|\mathbf{x}_t - \hat{\mathbf{x}}_t\|$ is upper-bounded by (68).

Decreasing the Step Size as $\sim 1/t$

Based on (68), the sequence of step sizes, $\{\eta_t\}$, may be designed to achieve some desired speed of convergence. An important example is when η_t decreases with t as

$$\eta_t = \frac{\eta_0}{t/\tau + 1},$$

for some positive constants η_0 and τ that determine the initial step size and the decay rate, respectively. In this case, the total error can be made to vanish as $\mathcal{O}(1/t)$.

Suppose η_0 is small enough that the calculations leading to (68) hold. Then for any non-negative i ,

$$\begin{aligned} \prod_{j=i}^{t-1} (1 - \eta_j \mu) &= \exp \left[\sum_{j=i}^{t-1} \ln \left(1 - \frac{\eta_0 \mu}{j/\tau + 1} \right) \right] \\ &\leq \exp \left[-\eta_0 \mu \tau \sum_{j=i}^{t-1} \frac{1}{j + \tau} \right] \\ &\leq \exp \left[-\eta_0 \mu \tau \int_{i+\tau}^{t+\tau} \frac{1}{z} dz \right] \\ &= \exp \left[-\eta_0 \mu \tau \ln \left(\frac{t+\tau}{i+\tau} \right) \right] = \left(\frac{i+\tau}{t+\tau} \right)^{\eta_0 \mu \tau}. \end{aligned} \quad (69)$$

The first term of (68) can be upper-bounded by using (69) with $i = 0$:

$$\prod_{i=0}^{t-1} (1 - \eta_i \mu) \leq \left(\frac{\tau}{t + \tau} \right)^{\eta_0 \mu \tau}. \quad (70)$$

$$\|\mathbf{x}_t - \hat{\mathbf{x}}_t\| \leq \left(\prod_{i=0}^{t-1} (1 - \eta_i \mu) \right) (\|\mathbf{x}_0 - \mathbf{x}_*\| + \|\hat{\mathbf{x}}_0 - \mathbf{x}_*\|) + 2 \frac{L}{\mu} \Lambda \sum_{i=0}^{t-1} |\eta_i - \eta_{i+1}| \prod_{j=i+1}^{t-1} (1 - \eta_j \mu) \cdot \|\nabla f(\mathbf{x}_*)\| \quad (68)$$

The second term of (68) can be upper-bounded, for $\tau \neq 1/(\eta_0 \mu)$, as follows:

$$\begin{aligned} & \sum_{i=0}^{t-1} (\eta_i - \eta_{i+1}) \prod_{j=i+1}^{t-1} (1 - \eta_j \mu) \\ & \stackrel{(a)}{\leq} \eta_0 \tau \sum_{i=0}^{t-1} \frac{1}{(i + \tau)(i + \tau + 1)} \left(\frac{i + 1 + \tau}{t + \tau} \right)^{\eta_0 \mu \tau} \\ & \leq \frac{\eta_0 \tau (1 + 1/\tau)}{(t + \tau)^{\eta_0 \mu \tau}} \sum_{i=0}^{t-1} (i + 1 + \tau)^{\eta_0 \mu \tau - 2} \\ & = \frac{\eta_0 (\tau + 1)}{(t + \tau)^{\eta_0 \mu \tau}} \sum_{i=1}^t (i + \tau)^{\eta_0 \mu \tau - 2} \\ & \stackrel{(b)}{\leq} \frac{\eta_0 (\tau + 1)}{(t + \tau)^{\eta_0 \mu \tau}} \int_0^{t+1} dz (z + \tau)^{\eta_0 \mu \tau - 2} \\ & = \frac{\eta_0 (\tau + 1)}{(t + \tau)^{\eta_0 \mu \tau}} \frac{(t + \tau + 1)^{\eta_0 \mu \tau - 1} - \tau^{\eta_0 \mu \tau - 1}}{\eta_0 \mu \tau - 1}, \quad (71) \end{aligned}$$

where in (a) we used (69). When upper-bounding the sum by an integral in (b), note that depending on $\eta_0 \mu \tau$, the terms in the sum may either be increasing or decreasing with i . Equations (70) and (71) together show that if $\tau > 1/(\eta_0 \mu)$, (68) approaches zero as $\mathcal{O}(1/t)$: the first term is $\mathcal{O}(1/t^{\eta_0 \mu \tau})$ and the second term is $\mathcal{O}(1/t)$. If $\tau < 1/(\eta_0 \mu)$, (68) approaches zero as $\mathcal{O}(1/t^{\eta_0 \mu \tau})$; in this case, both terms are of that order. At the “critical point” when $\tau = 1/(\eta_0 \mu)$, the last step in (71) does not hold but by evaluating the integral one can show that (68) decays as $\mathcal{O}(\log(t)/t)$.

A similar approach to the analysis of DGD with time-varying step sizes was used in [35].

V. Noisy DGD and Diffusion

The DGD and diffusion iterations in (38), and their counterparts with multiple local gradient updates [(52)–(53)] may be noisy. There are several possible origins of the noise: noise in the gradients due to random subsampling of data (stochastic DGD and diffusion); communication noise because of quantization or transmission errors when gradients are communicated between agents; or as an instance of the latter, randomness in the topology owing to, for example, communication link failures. Because of the noise, the iterates $\{\mathbf{x}_t\}$ become random variables. Next we analyze noisy DGD and diffusion in the MHT+contraction framework.

A. Data Subsampling: Stochastic DGD and Diffusion

The effect of random data subsampling can be modeled by adding noise onto the gradients. We consider first the case of a single local gradient update. Equation (38) becomes

$$\mathbf{x}_{t+1} = \mathbf{W} \mathbf{x}_t - \eta \mathbf{Z} (\nabla f(\mathbf{x}_t) + \boldsymbol{\epsilon}_t), \quad (72)$$

where $\boldsymbol{\epsilon}_t$ is the gradient noise at iteration t . Note that this noise is scaled by the step size, η .

The noise $\boldsymbol{\epsilon}_t$ is statistically dependent on $\mathbf{x}_t$, and potentially on other past iterates. However, if in every iteration the data are subsampled uniformly at random, $\boldsymbol{\epsilon}_t$ will have zero mean conditionally on $\mathbf{x}_t$,

$$\mathbb{E}\{\boldsymbol{\epsilon}_t | \mathbf{x}_t\} = \mathbf{0}. \quad (73)$$

We will assume that the conditional noise variance is at most affine in the norm-square of the iterates, so that

$$\mathbb{E}\{\|\eta \mathbf{Z} \boldsymbol{\epsilon}_t\|^2 | \mathbf{x}_t\} \leq \mathbb{E}\{\|\eta \boldsymbol{\epsilon}_t\|^2 | \mathbf{x}_t\} \leq \eta^2 (\sigma + \omega \|\mathbf{x}_t\|)^2, \quad (74)$$

for some non-negative constants σ and ω . Equation (74) is a standard model for sampling noise in stochastic gradient algorithms; for example, it is implied by [45, Eq. (48)] (see the remark at the end of Section II.B).³

Suppose η satisfies the condition (from the NC²T)

$$(1 - \eta \mu)^2 + \eta^2 \omega^2 < 1. \quad (75)$$

Applying the NC²T to the contractions defined in Section IV.C with $c = 1 - \eta \mu$, and σ and ω scaled by η , gives

$$\begin{aligned} & \sqrt{\mathbb{E}\{\|\mathbf{x}_t - \hat{\mathbf{x}}\|^2\}} \\ & \leq \hat{d} + \sqrt{(1 - \eta \mu)^2 + \eta^2 \omega^2}^t \left(\|\mathbf{x}_0 - \hat{\mathbf{x}}\| - \hat{d} \right)^+, \end{aligned}$$

where

$$\hat{d} = \frac{\eta \omega \|\hat{\mathbf{x}}\| + \eta \sigma}{\sqrt{1 - (1 - \eta \mu)^2} - \eta \omega} \stackrel{(a)}{\leq} \eta \frac{\omega \|\hat{\mathbf{x}} - \mathbf{x}_*\| + \omega \|\mathbf{x}_*\| + \sigma}{\sqrt{2\eta \mu - \eta^2 \mu^2} - \eta \omega}; \quad (76)$$

in (a) we used the triangle inequality. For small enough η ,

$$\frac{\eta}{\sqrt{2\eta \mu - \eta^2 \mu^2} - \eta \omega} \leq \frac{\sqrt{\eta}}{\sqrt{\mu}}, \quad (77)$$

which gives, from (76), the following simpler expression:

$$\hat{d} \leq \sqrt{\eta} \frac{\omega \|\hat{\mathbf{x}} - \mathbf{x}_*\| + \omega \|\mathbf{x}_*\| + \sigma}{\sqrt{\mu}}. \quad (78)$$

To bound the total error, we use (16):

$$\begin{aligned} \sqrt{\mathbb{E}\{\|\mathbf{x}_t - \mathbf{x}_*\|^2\}} & \leq \hat{d} + \|\hat{\mathbf{x}} - \mathbf{x}_*\| + \sqrt{(1 - \eta \mu)^2 + \eta^2 \omega^2}^t \\ & \quad \cdot \left(\|\mathbf{x}_0 - \mathbf{x}_*\| + \|\hat{\mathbf{x}} - \mathbf{x}_*\| - \hat{d} \right)^+. \quad (79) \end{aligned}$$

The distance $\|\hat{\mathbf{x}} - \mathbf{x}_*\|$ that enters into (79), both directly and via $\hat{d}$, is obtained from (47). Consider the three terms in (79). If η satisfies (77) then the contraction factor satisfies

$$\sqrt{(1 - \eta \mu)^2 + \eta^2 \omega^2} \leq \sqrt{1 - \eta \mu} \leq e^{-\eta \mu/2},$$

³As pointed out in [45], this condition is similar to [59, Eq. (4.3)]. While (74) and [45, Eq. (48)] express the upper bound on the conditional mean-square noise in terms of $\|\mathbf{x}_t\|$, [59, Eq. (4.3)] instead expresses this upper bound in terms of $\|\nabla f(\mathbf{x}_t)\|$; note that $\|\mathbf{x}_t\|$ and $\|\nabla f(\mathbf{x}_t)\|$ can be related to one another using the MHT.

and (75) in particular. For given $\epsilon > 0$, there exists η such that the first two terms in (79) sum up to less than $\epsilon/2$. For that η , there is a value of t such that the third term of (79) is less than $\epsilon/2$. In particular, η should be of order $\sim \epsilon^2$ to make the first two terms small, and t should be of order $\sim \log(1/\epsilon)/\epsilon^2$ to make the third term small.

1) The Impact of the Noise

In (79), $\|\hat{\mathbf{x}} - \mathbf{x}_*\| = \mathcal{O}(\eta)$, whereas $\hat{d} = \mathcal{O}(\sqrt{\eta})$. Hence, for small η , $\sqrt{\mathbb{E}\{\|\mathbf{x}_t - \mathbf{x}_*\|^2\}}$ is dominated by the effects of the gradient noisiness.

The noise has two effects. First, if $\omega > 0$, it slows down the convergence to the fixed point, $\hat{\mathbf{x}}$, by increasing the contraction factor from $1 - \eta\mu$ to $\sqrt{(1 - \eta\mu)^2 + \eta^2\omega^2}$, requiring a correspondingly smaller η . Second, the noise increases $\hat{d}$, by an amount dependent on both ω and σ . Note, however, that the fixed point itself is unaffected by the presence of noise.

2) Multiple Local Gradient Updates

To analyze convergence with multiple local gradient updates, we apply the NC³T to the contractions $\{\phi(\cdot)\}$ in (55)–(56). We obtain a bound with similar appearance as that in (79):

$$\sqrt{\mathbb{E}\{\|\mathbf{x}_t - \mathbf{x}_*\|^2\}} \leq \hat{d} + \|\hat{\mathbf{x}} - \mathbf{x}_*\| + \sqrt{(1 - \eta\mu)^2 + \eta^2\omega^2}^{tT} \cdot \left(\|\mathbf{x}_0 - \mathbf{x}_*\| + \|\hat{\mathbf{x}} - \mathbf{x}_*\| - \hat{d} \right)^+, \quad (80)$$

For small enough ηT ,

$$\hat{d} \leq \frac{\eta\omega M + \eta\sigma}{\sqrt{1 - (1 - \eta\mu)^2} - \eta\omega} \stackrel{(a)}{\leq} \sqrt{\eta} \frac{\omega\|\mathbf{x}_*\| + \sigma}{\sqrt{\mu}}, \quad (81)$$

where M quantifies how much the fixed point shifts because of the multiple local gradient updates; in (a), we used the bound on M from (116) in Appendix B:

$$M = \max\{\|\tilde{\mathbf{x}}_1\|, \dots, \|\tilde{\mathbf{x}}_T\|\} \leq \|\mathbf{x}_*\| + \mathcal{O}(\eta T),$$

where $\{\tilde{\mathbf{x}}_i\}$ are the auxiliary iterates defined in that appendix. Finally, in (80), $\|\hat{\mathbf{x}} - \mathbf{x}_*\|$ is bounded by (58) for DGD and (59) for diffusion.

B. Noisy Inter-Agent Communication

The communication between agents may be subject to errors caused by noise, interference, or signal quantization (e.g., sparsification). Such errors can be modeled by including a noise term ϵ_t that enters additively in (38):

$$\mathbf{x}_{t+1} = \mathbf{W}\mathbf{x}_t - \eta\mathbf{Z}\nabla f(\mathbf{x}_t) + \epsilon_t. \quad (82)$$

The analysis follows that for noisy gradients (Section V.A), but with an important difference: in the noisy-gradients case, ϵ_t is scaled by η ; here, it is not, since the noise affects the iterates communicated among the agents rather than the gradient updates.

We will assume that $\{\epsilon_t\}$ satisfy (73) and that

$$\mathbb{E}\{\|\epsilon_t\|^2 | \mathbf{x}_t\} \leq (\sigma + \omega\|\mathbf{x}_t\|)^2, \quad (83)$$

for some non-negative constants σ and ω . Equations (73) and (83) hold for example for noise and interference in analog transmission, and for quantization noise, if appropriate dithering is applied.

1) Scaling the Consensus Weights

There may not exist an η such that $(1 - \eta\mu)^2 + \omega^2 < 1$. Hence, the NC²T is not directly applicable; in fact, the iteration (82) may diverge. To guarantee convergence, we scale $\mathbf{W}$ using a *consensus step size*, γ , $0 < \gamma \leq 1$. With this modification, the noise-free iteration (38), becomes

$$\begin{aligned} \mathbf{x}_{t+1} &= (1 - \gamma)\mathbf{x}_t + \gamma\mathbf{W}\mathbf{x}_t - \eta\mathbf{Z}\nabla f(\mathbf{x}_t) \\ &= \mathbf{W}'\mathbf{x}_t - \eta\mathbf{Z}\nabla f(\mathbf{x}_t), \end{aligned}$$

where

$$\mathbf{W}' = (1 - \gamma)\mathbf{I} + \gamma\mathbf{W}. \quad (84)$$

(Note that $\mathbf{W}^T = \mathbf{W}'$.) The effect of the consensus step size scaling is to average the iterates over time, and the role of $1 - \gamma$ is similar to the forgetting factor in an autoregressive lowpass filter. Compared to the nominal $\mathbf{W}$, the effective weight matrix $\mathbf{W}'$ in (84) is pushed towards $\mathbf{I}$: the eigenvalues of $\mathbf{W}'$ are $\{1 - \gamma + \gamma\lambda_n\}$ and they satisfy (32) if the eigenvalues of $\mathbf{W}$, $\{\lambda_n\}$, do. For DGD, the requirement on the step size, (41), becomes

$$\eta \leq \frac{2 - \gamma + \gamma\lambda_N}{L + \mu},$$

while the requirement for diffusion is unchanged.

With the weight scaling, the noisy iteration (82) becomes

$$\mathbf{x}_{t+1} = \mathbf{W}'\mathbf{x}_t - \eta\mathbf{Z}\nabla f(\mathbf{x}_t) + \gamma\epsilon_t. \quad (85)$$

In particular, γ scales the communication noise in (85): by slowing down the consensus process, the noise is averaged over time, effectively scaling its variance by γ^2 .

2) Bounding the Error

Application of the NC²T with σ and ω scaled by γ , assuming that γ is small enough that $(1 - \eta\mu)^2 + \gamma^2\omega^2 < 1$, gives

$$\sqrt{\mathbb{E}\{\|\mathbf{x}_t - \mathbf{x}_*\|^2\}} \stackrel{(a)}{\leq} \hat{d} + \|\hat{\mathbf{x}} - \mathbf{x}_*\| + \sqrt{(1 - \eta\mu)^2 + \gamma^2\omega^2}^t \cdot \left(\|\mathbf{x}_0 - \mathbf{x}_*\| + \|\hat{\mathbf{x}} - \mathbf{x}_*\| - \hat{d} \right)^+, \quad (86)$$

where

$$\begin{aligned} \hat{d} &= \frac{\gamma\omega\|\hat{\mathbf{x}}\| + \gamma\sigma}{\sqrt{1 - (1 - \eta\mu)^2} - \gamma\omega} \\ &\leq \gamma \frac{\omega\|\hat{\mathbf{x}} - \mathbf{x}_*\| + \omega\|\mathbf{x}_*\| + \sigma}{\sqrt{2\eta\mu} - \eta^2\mu^2 - \gamma\omega} \\ &\stackrel{(a)}{\leq} \frac{\gamma}{\sqrt{\eta}} \frac{\omega\|\hat{\mathbf{x}} - \mathbf{x}_*\| + \omega\|\mathbf{x}_*\| + \sigma}{\sqrt{\mu}}, \end{aligned} \quad (87)$$

where (a) holds for small enough η and small enough $\gamma/\sqrt{\eta}$. The distance $\|\hat{x} - x_*\|$ that enters in (86) (directly and via $\hat{d}$) is bounded by (47), with W replaced by W' . In particular, λ_2 is replaced by the second-largest eigenvalue of W' , $1 - \gamma + \gamma\lambda_2$; hence, $1 - \lambda_2$ is replaced by $\gamma(1 - \lambda_2)$ in (51). In view of (47) and (51), this means that $\|\hat{x} - x_*\| = \mathcal{O}(\eta/\gamma)$. Hence η and γ must be jointly selected to make $\sqrt{E\{\|x_t - x_*\|^2\}}$ small. By taking $\gamma = \eta^{3/4}$, the first and the second terms in (86) are of the same order, $\sim \eta^{1/4}$. With this choice of γ , the contraction factor satisfies

$$\sqrt{(1 - \eta\mu)^2 + \eta^{3/2}\omega^2} \leq \sqrt{1 - \eta\mu} \leq e^{-\eta\mu/2},$$

for small enough η . For the so-selected, small enough, η and γ , the third term of (86) goes to 0 as $t \rightarrow \infty$.

A similar tuning of the learning (η) and consensus (γ) step sizes was suggested in [21] to mitigate quantization noise and in [35] to mitigate wireless propagation effects.

C. Random Topology

In some applications, the topology is random. This could happen if communication links between agents are activated (scheduled), or fail, at random. The iteration (38) then becomes

$$x_{t+1} = Q_t x_t - \eta Z_t \nabla f(x_t),$$

where Q_t is a (random) weight matrix that defines how, in iteration t , the agents aggregate the irregularly received updates from their neighbors, and Z_t is defined from Q_t similar to (37). To guarantee convergence, the consensus speed must be tuned, resulting in the modified iteration,

$$\begin{aligned} x_{t+1} &= (1 - \gamma)x_t + \gamma Q_t x_t - \eta Z_t \nabla f(x_t) \\ &= (1 - \gamma)x_t + \gamma E\{Q_t\} x_t - \eta E\{Z_t\} \nabla f(x_t) + \gamma \epsilon_t, \end{aligned} \quad (88)$$

where ϵ_t is effective noise:

$$\epsilon_t = (Q_t - E\{Q_t\})x_t - \frac{\eta}{\gamma}(Z_t - E\{Z_t\})\nabla f(x_t). \quad (89)$$

If Q_t is statistically independent of $\{x_1, \dots, x_t\}$ for all t then by inspection of (89), the condition (73) is satisfied. If $\{Q_t\}$ are identically distributed so that we can omit the subscript t , and have bounded second-order moments, then (83) is satisfied for some constant ω .

Equation (88) has the exact same form as (85), with W replaced by $E\{Q\}$, and consequently W' in (84) replaced by $(1 - \gamma)I + \gamma E\{Q\}$. Hence, the analysis in Section V.B applies, provided that $E\{Q\}$ satisfies the requirements made on W in Section IV.A. Note that we can have $Q^T \neq Q$ in general (hence, the instantaneous topology could be asymmetric and directed), but $E\{Q^T\} = E\{Q\}$ is required.

In case $\{Q_t\}$ are non-i.i.d., ϵ_t and $\epsilon_{t'}$ will become statistically dependent for $t \neq t'$, which causes (73) to be violated. An analysis of this case would require modeling of the joint statistics of $\{x_t\}$ over multiple iterations.

Algorithm	$T = 1$	$T > 1$
$\sqrt{E\{\ x_t - x_*\ ^2\}}$ w. sampl. noise [Eq. (72)]	(79) [$\hat{d}$ in (78)]	(80) [$\hat{d}$ in (81)]
$\sqrt{E\{\ x_t - x_*\ ^2\}}$ w. comm. noise [Eq. (85)]	(86) [$\hat{d}$ in (87)]	
$\ \hat{x} - x_*\ $ for DGD	(47) and (48)	(58)
$\ \hat{x} - x_*\ $ for diffusion	(47) and (48)	(59)

TABLE 2. Summary of the main convergence bounds for the noisy case.

D. Summary and Comments

Table 2 summarizes the bounds obtained for the noisy case.

Note how our MHT+contraction framework allows the conditional noise variance to scale with the squared magnitude of the iterates [see (74) and (83)]. In contrast, most literature (for example, [29]–[31], [54], [55], [60], [61]), assumes that the gradient noise variance is *uniformly bounded*, which corresponds to having $\omega = 0$ in (74) and (83). That assumption is violated in many problems of practical interest: for instance, as shown in Appendix C, it cannot hold for linear regression.

Examples of analyses in the literature that allow noise with the property (83), with $\omega > 0$, are [21], [22]. Theorem 2 in [21] also contains an error bound for noisy DGD with $\omega > 0$ (but $\sigma = 0$). This theorem only gives a scaling behavior; in contrast, our bounds are explicit.

VI. Application Example: DGD with Link Failures

As an application example, we consider standard DGD ($Z = I$) with inter-agent communication link failures. (Extensions to diffusion are possible, but require additional notation.) The literature on federated and decentralized learning with unreliable communications is quite rich; see Section I.A. Here we illustrate how the MHT+contraction framework, and the results in Sections V.B–C, can be directly used to analyze such setups.

Let W be a matrix of nominal weights associated with some given base topology, and denote by $p_{nn'}$ the probability that the transmission from agent n' to agent n succeeds. We assume that the success probabilities are positive and symmetric: $p_{nn'} = p_{n'n} > 0$ for all n and n' , and that all link failures are mutually independent. Given W , let S be a random matrix whose entries are independent and

$$S_{nn'} = \begin{cases} W_{nn'}, & \text{with probability } p_{nn'} \\ 0, & \text{with probability } 1 - p_{nn'} \end{cases}.$$

Replacing W in (35) with S does not guarantee convergence, but the use of the modified iteration (88) does, if Q is appropriately constructed based on S . Note that S is beyond control (it is determined by W and $\{p_{nn'}\}$), but Q can be *constructed* based on S , consistent with the topology defined by W – that is, allowing $Q_{nn'}$ to be nonzero only if

$W_{nn'}$ nonzero (for $n \neq n'$). Note also that while $\mathbf{W}^T = \mathbf{W}$, we have $\mathbf{S}^T \neq \mathbf{S}$ in general.

A. Known Link Success Probabilities $\{p_{nn'}\}$

If $\{p_{nn'}\}$ are known, we construct $\mathbf{Q}$ as follows:

$$Q_{nn'} = \begin{cases} 1 + S_{nn} - \sum_{n''=1}^N p_{nn''} W_{nn''}, & n = n' \\ S_{nn'}, & n \neq n', \end{cases} \quad (90)$$

Clearly,

$$\mathbb{E}\{Q_{nn'}\} = \begin{cases} 1 - \sum_{n''=1, n'' \neq n}^N p_{nn''} W_{nn''}, & n = n' \\ p_{nn'} W_{nn'}, & n \neq n'. \end{cases} \quad (91)$$

If the nominal weight matrix, $\mathbf{W}$, satisfies the conditions in Section IV.A then $\mathbb{E}\{\mathbf{Q}^T\} = \mathbb{E}\{\mathbf{Q}\}$, $\mathbb{E}\{\mathbf{Q}\}\mathbf{1} = \mathbf{1}$, and the graph induced by $\mathbb{E}\{\mathbf{Q}\}$ is connected.

For the analysis in Sections V.B–C to apply, we additionally require that the eigenvalues of $\mathbb{E}\{\mathbf{Q}\}$ satisfy (32). By the arguments in Section VII.E, a set of sufficient conditions for this is that $\mathbb{E}\{\mathbf{Q}\}$ is non-negative and not bipartite. These conditions in turn are satisfied if $\mathbf{W}$ is non-negative and not bipartite. (Note that, in contrast, non-negativity of $\mathbf{W}$ is not required by the main analysis in Sections IV and V.)

We have that $\mathbf{q}_n - \mathbb{E}\{\mathbf{q}_n\} = \mathbf{s}_n - \mathbb{E}\{\mathbf{s}_n\}$, where $\mathbf{s}_n^T$ is the n th row of $\mathbf{S}$ and $\mathbf{q}_n^T$ is the n th row of $\mathbf{Q}$. Also,

$$\text{Var}\{S_{nn'}\} = p_{nn'}(1 - p_{nn'})W_{nn'}^2.$$

The effective noise variance, computed via (89) with $\mathbf{Z}_t = \mathbf{I}$ and $\mathbf{Q}$ given by (90), can be upper-bounded as

$$\begin{aligned} \mathbb{E}\{\|\epsilon_t\|^2 | \mathbf{x}_t\} &= \mathbb{E}\left\{\sum_{n=1}^N [(\mathbf{q}_n - \mathbb{E}\{\mathbf{q}_n\})^T \mathbf{x}_t]^2 | \mathbf{x}_t\right\} \\ &\leq \sum_{n=1}^N \mathbb{E}\{\|\mathbf{q}_n - \mathbb{E}\{\mathbf{q}_n\}\|^2\} \cdot \|\mathbf{x}_t\|^2 \\ &= \sum_{n=1}^N \mathbb{E}\{\|\mathbf{s}_n - \mathbb{E}\{\mathbf{s}_n\}\|^2\} \cdot \|\mathbf{x}_t\|^2 \\ &= \sum_{n=1}^N \sum_{n'=1}^N p_{nn'}(1 - p_{nn'})W_{nn'}^2 \cdot \|\mathbf{x}_t\|^2 \\ &\stackrel{(a)}{\leq} \frac{1}{4} \text{Tr}\{\mathbf{W}^T \mathbf{W}\} \cdot \|\mathbf{x}_t\|^2 \\ &= \frac{1}{4} \sum_{n=1}^N \lambda_n^2 \|\mathbf{x}_t\|^2 \stackrel{(b)}{\leq} \frac{N}{4} \|\mathbf{x}_t\|^2, \end{aligned} \quad (92)$$

where in (a) we used that $x(1-x) \leq 1/4$ for all x , and in (b) we used that $\|\mathbf{W}\|_2 \leq 1$.

B. Unknown Link Success Probabilities $\{p_{nn'}\}$

If $\{p_{nn'}\}$ are unknown, $\mathbf{Q}$ in (90) is not computable. Instead, we may construct $\mathbf{Q}$ as follows:

$$Q_{nn'} = \begin{cases} 1 + S_{nn} - \sum_{n''=1}^N S_{nn''}, & n = n' \\ S_{nn'}, & n \neq n'. \end{cases}$$

Then (91) holds, and

$$\mathbf{q}_n - \mathbb{E}\{\mathbf{q}_n\} = (\mathbf{I} - \mathbf{e}_n \mathbf{1}^T)(\mathbf{s}_n - \mathbb{E}\{\mathbf{s}_n\}).$$

This gives the noise variance,

$$\begin{aligned} \mathbb{E}\{\|\epsilon_t\|^2 | \mathbf{x}_t\} &\leq \sum_{n=1}^N \mathbb{E}\{\|\mathbf{q}_n - \mathbb{E}\{\mathbf{q}_n\}\|^2\} \cdot \|\mathbf{x}_t\|^2 \\ &= \sum_{n=1}^N \text{Tr}\{(\mathbf{I} - \mathbf{e}_n \mathbf{1}^T) \text{Cov}\{\mathbf{s}_n\} (\mathbf{I} - \mathbf{1} \mathbf{e}_n^T)\} \cdot \|\mathbf{x}_t\|^2 \\ &\stackrel{(a)}{=} 2 \sum_{n=1}^N (\text{Tr}\{\text{Cov}\{\mathbf{s}_n\}\} - \text{Var}\{S_{nn}\}) \|\mathbf{x}_t\|^2 \\ &\leq \frac{1}{2} \sum_{n=1}^N \sum_{n'=1}^N W_{nn'}^2 \|\mathbf{x}_t\|^2 \leq \frac{N}{2} \|\mathbf{x}_t\|^2, \end{aligned} \quad (93)$$

where in (a) we used that the components of $\mathbf{s}_n$ are independent; hence $\text{Cov}\{\mathbf{s}_n\}$ is diagonal. Note how the noise variance doubles compared to the known-probabilities case.

C. Effects of $\{p_{nn'}\}$ on the Convergence Properties

The link success probabilities $\{p_{nn'}\}$ affect the distance in (47), which is determined by the second-largest eigenvalue of $\mathbb{E}\{\mathbf{Q}\}$ – along with, among others, the consensus step size γ . If $\mathbf{W}$ is non-negative (see the remark at the end of Section VI.A), a decrease in any of the link success probabilities pushes this eigenvalue towards 1. To see this, suppose we decrease $p_{ij} = p_{ji}$ by some positive δ , for some (i, j) , $i \neq j$. Let $\mathbb{E}\{\mathbf{Q}\}$ be $\mathbb{E}\{\mathbf{Q}\}$ after this decrease. This changes $\mathbb{E}\{Q_{ij}\}$ and $\mathbb{E}\{Q_{ji}\}$ by δW_{ij} , and $\mathbb{E}\{Q_{ii}\}$ and $\mathbb{E}\{Q_{jj}\}$ by $-\delta W_{ij}$. Hence,

$$\begin{aligned} \overline{\mathbb{E}\{\mathbf{Q}\}} - \mathbb{E}\{\mathbf{Q}\} &= \delta W_{ij}(\mathbf{e}_i \mathbf{e}_i^T + \mathbf{e}_j \mathbf{e}_j^T - \mathbf{e}_i \mathbf{e}_j^T - \mathbf{e}_j \mathbf{e}_i^T) \\ &= \delta W_{ij}(\mathbf{e}_i - \mathbf{e}_j)(\mathbf{e}_i - \mathbf{e}_j)^T \succcurlyeq \mathbf{0}, \end{aligned}$$

so the eigenvalues of $\mathbb{E}\{\mathbf{Q}\}$ increase (or stay unchanged). This effectively slows down the consensus speed.

The probabilities $\{p_{nn'}\}$ also affect the noise variance $\mathbb{E}\{\|\epsilon_t\|^2 | \mathbf{x}_t\}$. While the final upper bounds on this variance in (92) and (93) are not tight enough to show this dependence, inspection of the intermediate steps in (92) and (93) reveals that the variance increases if some p_{ij} is decreased, as long as $1/2 \leq p_{ij} \leq 1$.

VII. Additional Discussion, Insights, and Extensions

A. Vector-Valued Parameters

The MHT+contraction analysis approach extends directly to the case of vector-valued parameters. Nothing structurally changes in the problem and it is straightforward (but notationally cumbersome) to describe the DGD and diffusion iterations for the vector-valued case using a single equation. Let d be the dimension of the parameter vector, let $\mathbf{x}_{t,n}$ be the d -dimensional iterate held by agent n at iteration t , and define the $N \times d$ -matrix $\mathbf{X}_t = [\mathbf{x}_{t,1} \cdots \mathbf{x}_{t,N}]^T$. The local objectives become mappings $f_n : \mathbb{R}^d \rightarrow \mathbb{R}$, the global objective becomes a mapping $F : \mathbb{R}^d \rightarrow \mathbb{R}$, and the component-

wise objective becomes a mapping $f : \mathbb{R}^{N \times d} \rightarrow \mathbb{R}$. The counterpart of (38) becomes

$$\mathbf{X}_{t+1} = \mathbf{W}\mathbf{X}_t - \eta \mathbf{Z} \begin{bmatrix} [\nabla f_1(\mathbf{x}_{t,1})]^T \\ \vdots \\ [\nabla f_N(\mathbf{x}_{t,N})]^T \end{bmatrix}. \quad (94)$$

(Similar equations can be written up for the case with multiple local iterations.) Note how $\mathbf{W}$ acts on each column of $\mathbf{X}_t$ (data dimension) independently. To write (94) as a vector iteration, formally similar to (38), one may vectorize $\mathbf{X}_t$. For example, the first term of (94) can be written as

$$\text{vec}\{\mathbf{W}\mathbf{X}_t\} = (\mathbf{I} \otimes \mathbf{W})\text{vec}\{\mathbf{X}_t\}.$$

Note that $\mathbf{I} \otimes \mathbf{W}$ has the same eigenvalue spectrum as $\mathbf{W}$, but with every eigenvalue repeated d times.

B. Data Heterogeneity Metrics

In all bounds on $\|\hat{\mathbf{x}} - \mathbf{x}_*\|$, $\|\nabla f(\mathbf{x}_*)\|$ quantifies the heterogeneity among the objectives (data) held by the different agents. There are, at least, two alternative metrics that can be used to quantify this heterogeneity: the distance $\|\mathbf{x}_* - \mathbf{x}_{\text{loc}}^*\|$, and the objective function gap $f(\mathbf{x}_*) - f(\mathbf{x}_{\text{loc}}^*)$, where $\mathbf{x}_{\text{loc}}^* = [x_1^*, \dots, x_N^*]^T$ and x_n^* is the minimizer of the n th local objective, $f_n(\cdot)$.

The three different metrics $\|\nabla f(\mathbf{x}_*)\|$, $\|\mathbf{x}_* - \mathbf{x}_{\text{loc}}^*\|$, and $f(\mathbf{x}_*) - f(\mathbf{x}_{\text{loc}}^*)$ can be related to one another. Clearly, if all agents have the same objective, so that $\mathbf{x}_{\text{loc}}^* = \mathbf{x}_*$, all three metrics are zero. Since $\nabla f(\mathbf{x}_{\text{loc}}^*) = \mathbf{0}$, (18) and (19) give directly

$$\frac{1}{L} \|\nabla f(\mathbf{x}_*)\| \leq \|\mathbf{x}_* - \mathbf{x}_{\text{loc}}^*\| \leq \frac{1}{\mu} \|\nabla f(\mathbf{x}_*)\|.$$

Next, using Eq. (2.1.10) in [42, Th. 2.1.5] and Eq. (2.1.24) in [42, Th. 2.1.10], along with $\nabla f(\mathbf{x}_{\text{loc}}^*) = \mathbf{0}$, gives

$$\frac{1}{2L} \|\nabla f(\mathbf{x}_*)\|^2 \leq f(\mathbf{x}_*) - f(\mathbf{x}_{\text{loc}}^*) \leq \frac{1}{2\mu} \|\nabla f(\mathbf{x}_*)\|^2.$$

These inequalities show that if one of the heterogeneity metrics is small (or large), the others must also be small (or large). Note that, further upper-bounding $f(\mathbf{x}_*) - f(\mathbf{x}_{\text{loc}}^*) \leq f(\mathbf{0}) - f(\mathbf{x}_{\text{loc}}^*)$, with $\mathbf{0}$ interpreted as a common initialization, yields the data heterogeneity metric used in [7], [21] [$D^2/(2L)$ therein].

C. Multiple Consensus Rounds per Iteration

Multiple consensus averaging rounds can be performed before the local gradient update(s) are added, as suggested, for example in [15], [62]. With K consensus rounds using the same $\mathbf{W}$, the equivalent iteration becomes (38) with $\mathbf{W}$ replaced by $\mathbf{W}^K$. The second-largest eigenvalue of $\mathbf{W}^K$ becomes $\max\{\lambda_2^K, \lambda_N^K\}$, correspondingly affecting the convergence properties.

More generally, suppose that K consensus rounds are performed, with $\mathbf{W}^k$ as weight matrix in round k . Suppose, furthermore, that the agents store the consensus results after each round, $\{\mathbf{W}^k \mathbf{x}_t\}$, and form a weighted average thereof,

with non-negative weights $\{\alpha_k\}$ that satisfy $\sum_{k=1}^K \alpha_k = 1$. This results in the effective weight matrix

$$\mathbf{W}' = \sum_{k=1}^K \alpha_k \mathbf{W}^k.$$

As suggested for consensus problems in [63], $\{\alpha_k\}$ may be optimized. For example, one may minimize $\sum_{k=1}^K \alpha_k \max\{\lambda_2^k, \lambda_N^k\}$ with respect to $\{\alpha_k\}$. Alternatively, $\{\alpha_k\}$ could be chosen based on Chebyshev polynomials, as suggested by [64] – this is possible even if the eigenvalues of $\mathbf{W}$ are partly unknown. The number of consensus rounds, K , may be selected by considering the relative costs of communication and computation.

D. Breakdown of DGD for Bipartite Topologies

All analysis so far has assumed that (32) holds, and specifically that $\lambda_N > -1$.

For DGD, convergence breaks down if $\lambda_N = -1$. In this case, (41) cannot be satisfied as the numerator is zero. The case of $\lambda_N = -1$ can occur if the topology is bipartite. (For a bipartite graph, the eigenvalue spectrum is symmetric around 0, so $\lambda_N = -\lambda_1 = -1$.) For an example of a bipartite topology where the DGD iteration (35) diverges, take $N = 2$, $f_1(x) = f_2(x) = x^2/2$, and

$$\mathbf{W} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}.$$

Then $\lambda_1 = 1$ and $\lambda_2 = -1$. The matrix that appears in (40),

$$\mathbf{W} - \eta \mathbf{A}_{xy} = \begin{bmatrix} -\eta & 1 \\ 1 & -\eta \end{bmatrix}$$

has the eigenvalues $\pm 1 - \eta$, so $\phi(\cdot)$ is not a contraction for any η . In fact, with $\mathbf{x}_0 = [1 \ -1]^T$, $\|\mathbf{x}_t\| \rightarrow \infty$ as $t \rightarrow \infty$ in (35). A simple way to make DGD work for bipartite topologies is to tune the consensus speed (see Section V.B.1): provided that $\gamma < 1$, we have $1 - \gamma + \gamma\lambda_N > -1$ even if $\lambda_N = -1$.

For diffusion, (32) can be relaxed to permit $\lambda_N = -1$. With $\lambda_N = -1$, the diffusion iteration (36) converges; note, however, that $\mathbf{W}^t$ does not converge.

E. Sufficient Conditions for $\mathbf{W}$ to Satisfy (32)

A set of *sufficient* conditions for (32) to hold, in addition to the connectedness, $\mathbf{W}^T = \mathbf{W}$, and $\mathbf{W}\mathbf{1} = \mathbf{1}$ conditions, is that the elements of $\mathbf{W}$ are non-negative and that $\mathbf{W}$ is not bipartite. To see this, note that under these conditions, by the Perron-Frobenius theorem [65, Th. 8.4.4] there is a dominant eigenvector whose elements are strictly positive, and whose corresponding eigenvalue has multiplicity one and equals $\|\mathbf{W}\|_2$. Since the eigenvectors of $\mathbf{W}$ are mutually orthogonal, all eigenvectors but the dominant one have both positive and negative components. Therefore, the dominant eigenvector must be $\mathbf{1}$, with eigenvalue $\lambda_1 = 1$. Since this eigenvector has multiplicity one, and the eigenvalues are real-valued, $\lambda_2 < \lambda_1$. There could be more eigenvalues

Topology	$1 - \lambda_2$
Complete graph	$N\epsilon$
Star	ϵ
Line (path)	$4 \sin^2 \left(\frac{\pi}{2N} \right) \epsilon \approx \frac{\pi^2}{N^2} \epsilon$
Ring (closed path)	$4 \sin^2 \left(\frac{\pi}{N} \right) \epsilon \approx \frac{4\pi^2}{N^2} \epsilon$

TABLE 3. $1 - \lambda_2$ for some toy topologies with equal edge weights.

whose magnitudes equal $\|\mathbf{W}\|_2$, but the only possibility is -1 . Since $\mathbf{W}$ is not bipartite, it is primitive; then by [65, Th. 8.5.3], $\lambda_N > -1$. Hence, (32) is satisfied.

F. Equal Edge Weights

Additional insights into how the connectivity topology affects convergence can be gained by starting with an undirected, unweighted network and constructing $\mathbf{W}$ as follows:

$$\mathbf{W} = \mathbf{I} - \epsilon \mathbf{L}, \quad (95)$$

where $\mathbf{L}$ is the Laplacian of the network and ϵ is a positive constant. Then $\mathbf{W}^T = \mathbf{W}$, $\mathbf{W}\mathbf{1} = \mathbf{1}$, and all nonzero edge weights (excluding the loops) are equal to $-\epsilon$. Let $\{\Lambda_n\}$ be the eigenvalues of $\mathbf{L}$. Then

$$\begin{aligned} \lambda_n &= 1 - \epsilon \Lambda_n, \\ 0 &= \Lambda_1 < \Lambda_2 \leq \dots \leq \Lambda_N \stackrel{(a)}{\leq} 2k_{\max}, \end{aligned}$$

where $k_{\max}$ is the largest node degree and (a) follows from the Gershgorin disc theorem [65, Th. 6.1.1].⁴ Hence, if $\epsilon < 1/k_{\max}$, the eigenvalues of $\mathbf{W}$ in (95) satisfy (32).

Table 3 shows $1 - \lambda_2$ for equal edge weights and some toy topologies. The better connected the graph is (the smaller the diameter), the better is the convergence behavior.

VIII. Concluding Remarks

A rigorous analysis of the DGD and diffusion algorithms for differentiable, strongly convex and smooth objectives and symmetric weight matrices is advantageously approached by using our proposed MHT+contraction framework. Specifically, this independently yields, first, a tight contraction to a fixed point ($\hat{x}$), and second, a tight bound on the distance from $\hat{x}$ to the optimum (x_*). The resulting analyses extend directly to multiple local gradient updates, time-varying step sizes, sampling noise in the gradients, noisy communication links, and random topologies. In the noisy case, the approach avoids assumptions on bounded gradient noise variance.

Our analyses unify, considerably simplify, and extend results in the literature that were obtained using more complicated proof techniques. This makes our framework an attractive way of describing, using, and teaching decentralized convex optimization. Also, many of the presented results can be used, in a plug-and-play fashion, by practitioners who want to establish convergence of their algorithms without reproducing intricate technical arguments.

⁴A tighter inequality can be obtained, but is not of concern here.

The limitations of the MHT+contraction framework are: (1) the objectives must be (once) differentiable, strongly convex, and smooth; and (2) the weight matrix must be symmetric – except in the random-topology case, where it is enough that this matrix is symmetric *in expectation*. We stress that if the strong convexity assumption were dropped, the theory of contraction mappings would no longer apply. It remains an open problem whether our framework could be extended to handle deterministic asymmetric weight matrices.

Our bounds – similar to results in the literature – have the general form $c^t + \epsilon$, where c is a geometric factor and ϵ is an asymptotic error. We showed that the geometric factors (c) obtained in our analysis are the best possible. The bounds on the asymptotic error (ϵ) compare favorably with other known bounds (e.g., [7], [21]), but they could be improved upon further.

We believe that our proposed framework may be used to analyze other iterative algorithms for optimization over networks, such as gradient tracking. Another direction to explore is whether algorithms for optimization over nonlinear manifolds (e.g., [66]) could be cast in our framework.

REFERENCES

- [1] T. Yang, X. Yi, J. Wu, Y. Yuan, D. Wu, Z. Meng, Y. Hong, H. Wang, Z. Lin, and K.-H. Johansson, “A survey of distributed optimization,” *Annual Reviews in Control*, vol. 47, pp. 278–305, 2019.
- [2] T. T. Doan, S. T. Maguluri, and J. Romberg, “Convergence rates of distributed gradient methods under random quantization: a stochastic approximation approach,” *IEEE Transactions on Automatic Control*, vol. 66, no. 10, pp. 4469–4484, Oct. 2021.
- [3] S. Kar and J. M. Moura, “Consensus + innovations distributed inference over networks: cooperation and sensing in networked systems,” *IEEE Signal Processing Magazine*, vol. 30, no. 3, pp. 99–109, May 2013.
- [4] A. Nedić, J.-S. Pang, G. Scutari, and Y. Sun, *Multi-Agent Optimization*. Springer, Cham, 2018.
- [5] J. Tsitsiklis, D. Bertsekas, and M. Athans, “Distributed asynchronous deterministic and stochastic gradient optimization algorithms,” *IEEE Transactions on Automatic Control*, vol. 31, no. 9, pp. 803–812, Sep. 1986.
- [6] A. Nedić and A. Ozdaglar, “Distributed subgradient methods for multi-agent optimization,” *IEEE Transactions on Automatic Control*, vol. 54, no. 1, pp. 48–61, Jan. 2009.
- [7] K. Yuan, Q. Ling, and W. Yin, “On the convergence of decentralized gradient descent,” *SIAM Journal on Optimization*, vol. 26, no. 3, pp. 1835–1854, 2016.
- [8] K. Yuan, B. Ying, X. Zhao, and A. H. Sayed, “Exact diffusion for distributed optimization and learning – Part I: Algorithm development,” *IEEE Transactions on Signal Processing*, vol. 67, no. 3, pp. 708–723, Feb. 2019.
- [9] A. Nedić, A. Olshevsky, and W. Shi, “Achieving geometric convergence for distributed optimization over time-varying graphs,” *SIAM Journal on Optimization*, vol. 27, no. 4, pp. 2597–2633, 2017.
- [10] Y. Sun, G. Scutari, and A. Daneshmand, “Distributed optimization based on gradient tracking revisited: Enhancing convergence rate via surrogation,” *SIAM Journal on Optimization*, vol. 32, no. 2, pp. 354–385, 2022.
- [11] P. D. Lorenzo and G. Scutari, “Next: In-network nonconvex optimization,” *IEEE Transactions on Signal and Information Processing over Networks*, vol. 2, no. 2, pp. 120–136, Jun. 2016.
- [12] R. Xin, S. Pu, A. Nedić, and U. A. Khan, “A general framework for decentralized optimization with first-order methods,” *Proceedings of the IEEE*, vol. 108, no. 11, pp. 1869–1889, 2020.
- [13] B. McMahan, E. Moore, D. Ramage, S. Hampson, and B. A. y Arcas, “Communication-efficient learning of deep networks from decentral-

- ized data,” in *International Conference on Artificial Intelligence and Statistics*, 2017, pp. 1273–1282.
- [14] J. Xu, Y. Tian, Y. Sun, and G. Scutari, “Distributed algorithms for composite optimization: Unified framework and convergence analysis,” *IEEE Transactions on Signal Processing*, vol. 69, pp. 3555–3570, 2021.
 - [15] A. Koloskova, N. Loizou, S. Boreiri, M. Jaggi, and S. Stich, “A unified theory of decentralized SGD with changing topology and local updates,” in *International Conference on Machine Learning (ICML)*, 2020, pp. 5381–5393.
 - [16] J. Wang and G. Joshi, “Cooperative SGD: A unified framework for the design and analysis of local-update SGD algorithms,” *Journal of Machine Learning Research*, vol. 22, no. 213, pp. 1–50, 2021.
 - [17] A. Nedic, “Distributed gradient methods for convex machine learning problems in networks: Distributed optimization,” *IEEE Signal Processing Magazine*, vol. 37, no. 3, pp. 92–101, May 2020.
 - [18] R. Xin, S. Kar, and U. A. Khan, “Decentralized stochastic optimization and machine learning: A unified variance-reduction framework for robust performance and fast convergence,” *IEEE Signal Processing Magazine*, vol. 37, no. 3, pp. 102–113, May 2020.
 - [19] A. H. Sayed, *Inference and Learning from Data: Foundations*. Cambridge University Press, 2022, vol. 1.
 - [20] D. Jakovetic, D. Bajovic, A. K. Sahu, and S. Kar, “Convergence rates for distributed stochastic optimization over random networks,” in *IEEE Conference on Decision and Control (CDC)*, 2018, pp. 4238–4245.
 - [21] A. Reisizadeh, A. Mokhtari, H. Hassani, and R. Pedarsani, “An exact quantized decentralized gradient descent algorithm,” *IEEE Transactions on Signal Processing*, vol. 67, no. 19, pp. 4934–4947, Oct. 2019.
 - [22] N. Michelusi, G. Scutari, and C.-S. Lee, “Finite-bit quantization for distributed algorithms with linear convergence,” *IEEE Transactions on Information Theory*, vol. 68, no. 11, pp. 7254–7280, Nov. 2022.
 - [23] R. Saha, S. Rini, M. Rao, and A. J. Goldsmith, “Decentralized optimization over noisy, rate-constrained networks: Achieving consensus by communicating differences,” *IEEE Journal on Selected Areas in Communications*, vol. 40, no. 2, pp. 449–467, Feb. 2022.
 - [24] J. Wang, A. K. Sahu, G. Joshi, and S. Kar, “MATCHA: A matching-based link scheduling strategy to speed up distributed optimization,” *IEEE Transactions on Signal Processing*, vol. 70, pp. 5208–5221, 2022.
 - [25] C.-C. Chiu, X. Zhang, T. He, S. Wang, and A. Swami, “Laplacian matrix sampling for communication-efficient decentralized learning,” *IEEE Journal on Selected Areas in Communications*, vol. 41, no. 4, pp. 887–901, Apr. 2023.
 - [26] Y. Mao, Z. Zhao, M. Yang, L. Liang, Y. Liu, W. Ding, T. Lan, and X.-P. Zhang, “SAFARI: Sparsity-enabled federated learning with limited and unreliable communications,” *IEEE Transactions on Mobile Computing*, vol. 23, no. 5, pp. 4819–4831, May 2024.
 - [27] A. Rodio, G. Neglia, F. Busacca, S. Mangione, S. Palazzo, F. Restuccia, and I. Tinnirello, “Federated learning with packet losses,” in *International Symposium on Wireless Personal Multimedia Communications (WPMC)*, 2023.
 - [28] M. M. Amiri and D. Gündüz, “Federated learning over wireless fading channels,” *IEEE Transactions on Wireless Communications*, vol. 19, no. 5, pp. 3546–3557, May 2020.
 - [29] G. Zhu, Y. Wang, and K. Huang, “Broadband analog aggregation for low-latency federated edge learning,” *IEEE Transactions on Wireless Communications*, vol. 19, no. 1, pp. 491–506, Jan. 2020.
 - [30] M. M. Amiri, D. Gündüz, S. R. Kulkarni, and H. V. Poor, “Convergence of federated learning over a noisy downlink,” *IEEE Transactions on Wireless Communications*, vol. 21, no. 3, pp. 1422–1437, Mar. 2022.
 - [31] H. Xing, O. Simeone, and S. Bi, “Federated learning over wireless device-to-device networks: Algorithms and convergence analysis,” *IEEE Journal on Selected Areas in Communications*, vol. 39, no. 12, pp. 3723–3741, Dec. 2021.
 - [32] Y. Shi, Y. Zhou, and Y. Shi, “Over-the-air decentralized federated learning,” in *IEEE International Symposium on Information Theory (ISIT)*, 2021, pp. 455–460.
 - [33] E. Ozfatura, S. Rini, and D. Gündüz, “Decentralized SGD with over-the-air computation,” in *IEEE Global Communications Conference (GLOBECOM)*, 2020.
 - [34] H. Ye, L. Liang, and G. Y. Li, “Decentralized federated learning with unreliable communications,” *IEEE Journal of Selected Topics in Signal Processing*, vol. 16, no. 3, pp. 487–500, Apr. 2022.
 - [35] N. Michelusi, “Non-coherent over-the-air decentralized gradient descent,” *IEEE Transactions on Signal Processing*, vol. 72, pp. 4618–4634, 2024.
 - [36] Z. Jiang, G. Yu, Y. Cai, and Y. Jiang, “Decentralized edge learning via unreliable device-to-device communications,” *IEEE Transactions on Wireless Communications*, vol. 21, no. 11, pp. 9041–9055, Nov. 2022.
 - [37] P. S. Bouzinis, P. D. Diamantoulakis, and G. K. Karagiannidis, “Wireless federated learning (WFL) for 6G networks. Part I: Research challenges and future trends,” *IEEE Communications Letters*, vol. 26, no. 1, pp. 3–7, Jan. 2022.
 - [38] M. Beitollahi and N. Lu, “Federated learning over wireless networks: Challenges and solutions,” *IEEE Internet of Things Journal*, vol. 10, no. 16, pp. 14 749–14 763, Aug. 2023.
 - [39] E. T. Martínez Beltrán, M. Q. Pérez, P. M. S. Sánchez, S. L. Bernal, G. Bovet, M. G. Pérez, G. M. Pérez, and A. H. Celdrán, “Decentralized federated learning: Fundamentals, state of the art, frameworks, trends, and challenges,” *IEEE Communications Surveys & Tutorials*, vol. 25, no. 4, pp. 2983–3013, 2023.
 - [40] M. Chen, D. Gündüz, K. Huang, W. Saad, M. Bennis, A. V. Feljan, and H. V. Poor, “Distributed learning in wireless networks: Recent progress and future challenges,” *IEEE Journal on Selected Areas in Communications*, vol. 39, no. 12, pp. 3579–3605, Dec. 2021.
 - [41] T. Gafni, N. Shlezinger, K. Cohen, Y. C. Eldar, and H. V. Poor, “Federated learning: A signal processing perspective,” *IEEE Signal Processing Magazine*, vol. 39, no. 3, pp. 14–41, May 2022.
 - [42] Y. Nesterov, *Lectures on Convex Optimization*. Springer, 2nd edition, 2018.
 - [43] A. H. Sayed, “Adaptive networks,” *Proceedings of the IEEE*, vol. 102, no. 4, pp. 460–497, Apr. 2014.
 - [44] R. Nassif, S. Vlaski, and A. H. Sayed, “Adaptation and learning over networks under subspace constraints – Part I: Stability analysis,” *IEEE Transactions on Signal Processing*, vol. 68, pp. 1346–1360, 2020.
 - [45] J. Chen and A. H. Sayed, “Diffusion adaptation strategies for distributed optimization and learning over networks,” *IEEE Transactions on Signal Processing*, vol. 60, no. 8, pp. 4289–4305, 2012.
 - [46] W. Rudin, *Principles of Mathematical Analysis*. McGraw-Hill, 1953.
 - [47] Z. Yang, M. Chen, W. Saad, C. S. Hong, and M. Shikh-Bahaei, “Energy efficient federated learning over wireless communication networks,” *IEEE Transactions on Wireless Communications*, vol. 20, no. 3, pp. 1935–1949, Mar. 2021.
 - [48] C. Feng, H. Wang, Y. Han, Y. Xia, and X. M. Tu, “The mean value theorem and Taylor’s expansion in statistics,” *The American Statistician*, vol. 67, no. 4, pp. 245–248, 2013.
 - [49] S. Bubeck, “Convex optimization: Algorithms and complexity,” *Foundations and Trends® in Machine Learning*, vol. 8, no. 3–4, pp. 231–357, 2015.
 - [50] B. T. Polyak, *Introduction to Optimization*. Optimization Software, 1987.
 - [51] R. A. Polyak, *Introduction to Continuous Optimization*. Springer, 2021.
 - [52] S. Shalev-Shwartz and S. Ben-David, *Understanding Machine Learning: From Theory to Algorithms*. Cambridge University Press, 2014.
 - [53] X. Li, K. Huang, W. Yang, S. Wang, and Z. Zhang, “On the convergence of FedAvg on non-i.i.d. data,” in *International Conference on Learning Representations (ICLR)*, 2020.
 - [54] S. U. Stich, “Local SGD converges fast and communicates little,” in *International Conference on Learning Representations (ICLR)*, 2019.
 - [55] Y. Ruan, X. Zhang, S.-C. Liang, and C. Joe-Wong, “Towards flexible device participation in federated learning,” in *International Conference on Artificial Intelligence and Statistics*, 2021, pp. 3403–3411.
 - [56] A. Jadbabaie, A. Makur, and D. Shah, “Federated optimization of smooth loss functions,” *IEEE Transactions on Information Theory*, vol. 69, no. 12, pp. 7836–7866, Dec. 2023.
 - [57] L. M. Nguyen, P. H. Nguyen, P. Richtárik, K. Scheinberg, M. Takáč, and M. van Dijk, “New convergence aspects of stochastic gradient algorithms,” *Journal of Machine Learning Research*, vol. 20, no. 176, pp. 1–49, 2019.
 - [58] A. H. Sayed, “Diffusion adaptation over networks,” in *Academic Press Library in Signal Processing: Volume 3*, A. M. Zoubir, M. Viberg, R. Chellappa, and S. Theodoridis, Eds. Elsevier, 2014, vol. 3, pp. 323–453.

- [59] D. P. Bertsekas and J. N. Tsitsiklis, "Gradient convergence in gradient methods with errors," *SIAM Journal on Optimization*, vol. 10, no. 3, pp. 627–642, 2000.
- [60] X. Wei and C. Shen, "Federated learning over noisy channels: Convergence analysis and design examples," *IEEE Transactions on Cognitive Communications and Networking*, vol. 8, no. 2, pp. 1253–1268, Jun. 2022.
- [61] F. P.-C. Lin, S. Hosseinalipour, S. S. Azam, C. G. Brinton, and N. Michelusi, "Semi-decentralized federated learning with cooperative D2D local model aggregations," *IEEE Journal on Selected Areas in Communications*, vol. 39, no. 12, pp. 3851–3869, Dec. 2021.
- [62] A. S. Berahas, R. Bollapragada, N. S. Keskar, and E. Wei, "Balancing communication and computation in distributed optimization," *IEEE Transactions on Automatic Control*, vol. 64, no. 8, pp. 3141–3155, 2019.
- [63] R. Berthier, F. Bach, and P. Gaillard, "Accelerated gossip in networks of given dimension using Jacobi polynomial iterations," *SIAM Journal on Mathematics of Data Science*, vol. 2, no. 1, pp. 24–47, 2020.
- [64] K. Scaman, F. Bach, S. Bubeck, Y. T. Lee, and L. Massoulié, "Optimal algorithms for smooth and strongly convex distributed optimization in networks," in *International Conference on Machine Learning (ICML)*, 2017, pp. 3027–3036.
- [65] R. A. Horn and C. R. Johnson, *Matrix Analysis*. Cambridge University Press, 1999.
- [66] S. Chen, A. Garcia, M. Hong, and S. Shahrampour, "Decentralized Riemannian gradient descent on the Stiefel manifold," in *International Conference on Machine Learning (ICML)*, 2021, pp. 1594–1605.

Appendix

A. An Auxiliary Result on Distance Bounding

Consider the solution with respect to $\mathbf{d}$ of the linear equation

$$\left(\mathbf{Z}\mathbf{A} + \frac{1}{\eta}(\mathbf{I} - \mathbf{W}) \right) \mathbf{d} = -\mathbf{Z}\mathbf{v}, \quad (96)$$

where $\mathbf{Z}$ is defined in (37), $L\mathbf{I} \succcurlyeq \mathbf{A} \succcurlyeq \mu\mathbf{I}$, $\mathbf{W}$ satisfies the requirements set forth in Section IV.A, and the vector $\mathbf{v}$ satisfies $\Pi_1 \mathbf{v} = \mathbf{0}$. Then

$$\|\mathbf{d}\| \leq \eta \frac{L}{\mu} \Lambda \|\mathbf{v}\|, \quad (97)$$

where Λ is defined in (48), and where for diffusion we also require $\eta \leq 1/(L\Lambda)$.

Proof of (97)

Multiplying (96) from the left by Π_1 and using $\Pi_1(\mathbf{I} - \mathbf{W}) = \mathbf{0}$, in either case, yields $\Pi_1 \mathbf{A} \mathbf{d} = \mathbf{0}$. Then

$$\mathbf{0} = \Pi_1 \mathbf{A} (\Pi_1 \mathbf{d} + \Pi_1^\perp \mathbf{d}) = \frac{\mathbf{1}^T \mathbf{A} \mathbf{1}}{N} \Pi_1 \mathbf{d} + \Pi_1 \mathbf{A} \Pi_1^\perp \mathbf{d} = \frac{\mathbf{1}^T \mathbf{A} \mathbf{1}}{N} (\mathbf{d} - \Pi_1^\perp \mathbf{d}) + \Pi_1 \mathbf{A} \Pi_1^\perp \mathbf{d},$$

which shows that

$$\mathbf{d} = \left(\mathbf{I} - \frac{N}{\mathbf{1}^T \mathbf{A} \mathbf{1}} \Pi_1 \mathbf{A} \right) \Pi_1^\perp \mathbf{d} = \left(\mathbf{A}^{-1} - \frac{N}{\mathbf{1}^T \mathbf{A} \mathbf{1}} \Pi_1 \right) \mathbf{A} \Pi_1^\perp \mathbf{d}. \quad (98)$$

Note that the matrix inside the second parenthesis in (98) satisfies

$$\frac{1}{\mu} \mathbf{I} \succcurlyeq \mathbf{A}^{-1} - \frac{N}{\mathbf{1}^T \mathbf{A} \mathbf{1}} \Pi_1 \succcurlyeq \mathbf{0}, \quad (99)$$

where the second term has only one nonzero eigenvalue, which is at most $1/\mu$ because $\mathbf{1}^T \mathbf{A} \mathbf{1} \geq N\mu$. Then (98) implies

$$\|\mathbf{d}\| \leq \frac{L}{\mu} \|\Pi_1^\perp \mathbf{d}\|. \quad (100)$$

If $\Pi_1^\perp \mathbf{d} = \mathbf{0}$ there is nothing to show; hence we assume that $\Pi_1^\perp \mathbf{d} \neq \mathbf{0}$.

- For DGD ($\mathbf{Z} = \mathbf{I}$), multiply (96) from the left by $\mathbf{d}^T \Pi_1^\perp$ and insert (98); using $(\mathbf{I} - \mathbf{W}) \Pi_1 = \mathbf{0}$ this gives

$$\mathbf{d}^T \Pi_1^\perp \left[\mathbf{A} \left(\mathbf{A}^{-1} - \frac{N}{\mathbf{1}^T \mathbf{A} \mathbf{1}} \Pi_1 \right) \mathbf{A} + \frac{1}{\eta} (\mathbf{I} - \mathbf{W}) \right] \Pi_1^\perp \mathbf{d} = -\mathbf{d}^T \Pi_1^\perp \mathbf{v}. \quad (101)$$

Using (99) and the fact that $\mathbf{I} - \mathbf{W} \succcurlyeq (1 - \lambda_2) \Pi_1^\perp$, along with Cauchy-Schwarz inequality, (101) implies that

$$\frac{1 - \lambda_2}{\eta} \|\Pi_1^\perp \mathbf{d}\|^2 \leq \|\Pi_1^\perp \mathbf{d}\| \cdot \|\mathbf{v}\|,$$

that is,

$$\|\Pi_1^\perp \mathbf{d}\| \leq \frac{\eta}{1 - \lambda_2} \|\mathbf{v}\|. \quad (102)$$

Equation (102), combined with (100), gives the expression for DGD in (97).

- For diffusion ($\mathbf{Z} = \mathbf{W}$), instead multiply (96) from the left by $\mathbf{d}^T \Pi_1^\perp (\mathbf{I} - \mathbf{W})^\dagger$. Using that $(\mathbf{I} - \mathbf{W})^\dagger (\mathbf{I} - \mathbf{W}) = \Pi_1^\perp$ (since $\lambda_2 < 1$), this gives

$$\mathbf{d}^T \Pi_1^\perp (\mathbf{I} - \mathbf{W})^\dagger \mathbf{W} \mathbf{A} \mathbf{d} + \frac{1}{\eta} \mathbf{d}^T \Pi_1^\perp \mathbf{d} = -\mathbf{d}^T \Pi_1^\perp (\mathbf{I} - \mathbf{W})^\dagger \mathbf{W} \mathbf{v}. \quad (103)$$

Applying Cauchy-Schwarz inequality to (103) gives

$$\frac{1}{\eta} \|\Pi_1^\perp \mathbf{d}\|^2 \leq \|\Pi_1^\perp \mathbf{d}\| \cdot \|(\mathbf{I} - \mathbf{W})^\dagger \mathbf{W} \mathbf{v}\| + \|\Pi_1^\perp \mathbf{d}\| \cdot \|(\mathbf{I} - \mathbf{W})^\dagger \mathbf{W} \mathbf{A} \mathbf{d}\|.$$

Dividing both sides of (103) by $\|\Pi_1^\perp \mathbf{d}\|$ and inserting (98) yields

$$\begin{aligned} \frac{1}{\eta} \|\Pi_1^\perp \mathbf{d}\| &\leq \|(\mathbf{I} - \mathbf{W})^\dagger \mathbf{W} \mathbf{v}\| + \left\| (\mathbf{I} - \mathbf{W})^\dagger \mathbf{W} \mathbf{A} \left(\mathbf{A}^{-1} - \frac{N}{\mathbf{1}^\top \mathbf{A} \mathbf{1}} \Pi_1 \right) \mathbf{A} \Pi_1^\perp \mathbf{d} \right\| \\ &\leq \|(\mathbf{I} - \mathbf{W})^\dagger \mathbf{W} \mathbf{v}\| + \|(\mathbf{I} - \mathbf{W})^\dagger \mathbf{W}\|_2 \left\| \mathbf{A} \left(\mathbf{A}^{-1} - \frac{N}{\mathbf{1}^\top \mathbf{A} \mathbf{1}} \Pi_1 \right) \mathbf{A} \right\|_2 \|\Pi_1^\perp \mathbf{d}\| \\ &\stackrel{(a)}{\leq} \|(\mathbf{I} - \mathbf{W})^\dagger \mathbf{W} \mathbf{v}\| + L \|(\mathbf{I} - \mathbf{W})^\dagger \mathbf{W}\|_2 \|\Pi_1^\perp \mathbf{d}\|, \end{aligned}$$

where in (a) we used that (99) implies $\mathbf{A}(\mathbf{A}^{-1} - \frac{N}{\mathbf{1}^\top \mathbf{A} \mathbf{1}} \Pi_1) \mathbf{A} \preceq L \mathbf{I}$. Hence,

$$\|\Pi_1^\perp \mathbf{d}\| \leq \frac{\eta}{1 - \eta L \|(\mathbf{I} - \mathbf{W})^\dagger \mathbf{W}\|_2} \|(\mathbf{I} - \mathbf{W})^\dagger \mathbf{W} \mathbf{v}\| \stackrel{(a)}{\leq} 2\eta \|(\mathbf{I} - \mathbf{W})^\dagger \mathbf{W} \mathbf{v}\| \leq 2\eta \|(\mathbf{I} - \mathbf{W})^\dagger \mathbf{W}\|_2 \|\mathbf{v}\|,$$

where (a) holds if $\eta \leq 1/(L\Lambda)$.⁵ Using again (100) we get the formula for diffusion in (97). $\blacksquare$

B. Derivation of (58) and (59)

The strategy is to obtain a linear equation that relates $\hat{\mathbf{x}} - \mathbf{x}_*$ to $\nabla f(\mathbf{x}_*)$, where $\hat{\mathbf{x}}$ is the fixed point of $\varphi(\mathbf{x})$, and then from this equation upper-bound $\|\hat{\mathbf{x}} - \mathbf{x}_*\|$.

We first show the following auxiliary equality:

$$(\mathbf{I} - \eta \mathbf{A}_{T-2}) \cdots (\mathbf{I} - \eta \mathbf{A}_i) = \mathbf{I} - \eta \sum_{k=i}^{T-2} \mathbf{A}_k + \mathcal{O}(\eta^2 T^2), \quad i = 0, \dots, T-2, \quad (104)$$

where $\{\mathbf{A}_i\}$ is a sequence of matrices such that $\mu \mathbf{I} \preceq \mathbf{A}_i \preceq L \mathbf{I}$. To establish (104), define, for $i = 0, \dots, T-2$,

$$\mathbf{E}_i = (\mathbf{I} - \eta \mathbf{A}_{T-2}) \cdots (\mathbf{I} - \eta \mathbf{A}_i) - \left(\mathbf{I} - \eta \sum_{k=i}^{T-2} \mathbf{A}_k \right).$$

Clearly, (104) holds if

$$\|\mathbf{E}_i\|_2 \leq \eta^2 (T-2-i)^2 L^2, \quad (105)$$

for $i = 0, \dots, T-2$. Equation (105) holds for $i = T-2$, since $\mathbf{E}_{T-2} = \mathbf{0}$. Suppose, as induction hypothesis, that (105) is true for some i . Then

$$\begin{aligned} \|\mathbf{E}_{i-1}\|_2 &= \left\| (\mathbf{I} - \eta \mathbf{A}_{T-2}) \cdots (\mathbf{I} - \eta \mathbf{A}_{i-1}) - \left(\mathbf{I} - \eta \sum_{k=i-1}^{T-2} \mathbf{A}_k \right) \right\|_2 \\ &= \left\| \left[\mathbf{E}_i + \left(\mathbf{I} - \eta \sum_{k=i}^{T-2} \mathbf{A}_k \right) \right] (\mathbf{I} - \eta \mathbf{A}_{i-1}) - \left(\mathbf{I} - \eta \sum_{k=i-1}^{T-2} \mathbf{A}_k \right) \right\|_2 \\ &= \left\| \mathbf{E}_i (\mathbf{I} - \eta \mathbf{A}_{i-1}) + \eta^2 \sum_{k=i}^{T-2} \mathbf{A}_k \mathbf{A}_{i-1} \right\|_2 \stackrel{(a)}{\leq} \|\mathbf{E}_i\|_2 + \eta^2 \sum_{k=i}^{T-2} \|\mathbf{A}_k \mathbf{A}_{i-1}\|_2 \\ &\leq \eta^2 (T-2-i)^2 L^2 + \eta^2 (T-1-i) L^2 \leq \eta^2 (T-1-i)^2 L^2, \end{aligned}$$

where in (a) we used that $\|\mathbf{I} - \eta \mathbf{A}_{i-1}\|_2 \leq 1$. Hence, (105) holds for $i-1$. By induction over i , (105), and therefore (104), hold for all i .

Now, to relate $\hat{\mathbf{x}} - \mathbf{x}_*$ to $\nabla f(\mathbf{x}_*)$, first introduce the following auxiliary variables $\tilde{\mathbf{x}}_0, \dots, \tilde{\mathbf{x}}_T$ to describe the mapping $\varphi(\mathbf{x})$ ($\tilde{\mathbf{x}}_i$ represents the result after the i th local gradient update):

$$\begin{aligned} \tilde{\mathbf{x}}_0 &= \mathbf{x}, \\ \tilde{\mathbf{x}}_{i+1} &= \phi_i(\tilde{\mathbf{x}}_i) = \tilde{\mathbf{x}}_i - \eta \nabla f(\tilde{\mathbf{x}}_i), \quad i = 0, \dots, T-2 \\ \tilde{\mathbf{x}}_T &= \varphi(\mathbf{x}) = \phi_T(\tilde{\mathbf{x}}_{T-1}) = \mathbf{W} \tilde{\mathbf{x}}_{T-1} - \eta \mathbf{Z} \nabla f(\tilde{\mathbf{x}}_{T-1}), \end{aligned} \quad (106)$$

where $\mathbf{Z}$ is defined in (37). Note that when $\varphi(\mathbf{x})$ is applied to its fixed point, $\hat{\mathbf{x}}$, it holds that

$$\tilde{\mathbf{x}}_0 = \tilde{\mathbf{x}}_T = \varphi(\hat{\mathbf{x}}) = \hat{\mathbf{x}}.$$

⁵This bound can be tightened slightly by enforcing a stricter condition on η . For an arbitrary $\alpha > 1$, we have $1/(1 - \eta L \|(\mathbf{I} - \mathbf{W})^\dagger \mathbf{W}\|_2) \leq \alpha$ if $\eta \leq (\alpha - 1)/(\alpha L \|(\mathbf{I} - \mathbf{W})^\dagger \mathbf{W}\|_2)$. We took $\alpha = 2$, which results in the cleanest-looking, but not tightest possible, bound.

From (106), we obtain

$$\begin{aligned}\tilde{\mathbf{x}}_0 - \mathbf{x}_* &= \hat{\mathbf{x}} - \mathbf{x}_*, \\ \tilde{\mathbf{x}}_{i+1} - \mathbf{x}_* &= \tilde{\mathbf{x}}_i - \eta \nabla f(\tilde{\mathbf{x}}_i) - \mathbf{x}_* \\ &\stackrel{(a)}{=} \tilde{\mathbf{x}}_i - \eta [\nabla f(\mathbf{x}_*) + \mathbf{A}_i(\tilde{\mathbf{x}}_i - \mathbf{x}_*)] - \mathbf{x}_* \\ &= (\mathbf{I} - \eta \mathbf{A}_i)(\tilde{\mathbf{x}}_i - \mathbf{x}_*) - \eta \nabla f(\mathbf{x}_*), \quad i = 0, \dots, T-2,\end{aligned}$$

where in (a) we applied the MHT to $f(\cdot)$ at the points $\mathbf{x}_*$ and $\tilde{\mathbf{x}}_i$, and used $\mathbf{A}_i$ as a shorthand for the kernel $\mathbf{A}_{\mathbf{x}_*, \tilde{\mathbf{x}}_i}$. By induction over i , and using (104),

$$\begin{aligned}\tilde{\mathbf{x}}_{T-1} - \mathbf{x}_* &= (\mathbf{I} - \eta \mathbf{A}_{T-2}) \cdots (\mathbf{I} - \eta \mathbf{A}_0)(\tilde{\mathbf{x}}_0 - \mathbf{x}_*) - \eta \sum_{i=1}^{T-1} (\mathbf{I} - \eta \mathbf{A}_{T-2}) \cdots (\mathbf{I} - \eta \mathbf{A}_i) \nabla f(\mathbf{x}_*) \\ &= \left(\mathbf{I} - \eta \sum_{i=0}^{T-2} \mathbf{A}_i + \mathcal{O}(\eta^2 T^2) \right) (\tilde{\mathbf{x}}_0 - \mathbf{x}_*) - \eta \left((T-1) \mathbf{I} - \eta \sum_{i=1}^{T-1} \sum_{j=i}^{T-2} \mathbf{A}_j + \mathcal{O}(\eta^2 T^3) \right) \nabla f(\mathbf{x}_*). \quad (107)\end{aligned}$$

Also,

$$\begin{aligned}\hat{\mathbf{x}} - \mathbf{x}_* &= \tilde{\mathbf{x}}_T - \mathbf{x}_* \\ &= \mathbf{W} \tilde{\mathbf{x}}_{T-1} - \eta \mathbf{Z} \nabla f(\tilde{\mathbf{x}}_{T-1}) - \mathbf{x}_* \\ &= \mathbf{W}(\tilde{\mathbf{x}}_{T-1} - \mathbf{x}_*) - \eta \mathbf{Z} \nabla f(\tilde{\mathbf{x}}_{T-1}) \\ &\stackrel{(a)}{=} \mathbf{W}(\tilde{\mathbf{x}}_{T-1} - \mathbf{x}_*) - \eta \mathbf{Z} [\nabla f(\mathbf{x}_*) + \mathbf{A}_{T-1}(\tilde{\mathbf{x}}_{T-1} - \mathbf{x}_*)] \\ &= (\mathbf{W} - \eta \mathbf{Z} \mathbf{A}_{T-1})(\tilde{\mathbf{x}}_{T-1} - \mathbf{x}_*) - \eta \mathbf{Z} \nabla f(\mathbf{x}_*) \\ &\stackrel{(b)}{=} (\mathbf{W} - \eta \mathbf{Z} \mathbf{A}_{T-1}) \left[\left(\mathbf{I} - \eta \sum_{i=0}^{T-2} \mathbf{A}_i + \mathcal{O}(\eta^2 T^2) \right) (\tilde{\mathbf{x}}_0 - \mathbf{x}_*) - \eta \left((T-1) \mathbf{I} - \eta \sum_{i=1}^{T-1} \sum_{j=i}^{T-2} \mathbf{A}_j + \mathcal{O}(\eta^2 T^3) \right) \nabla f(\mathbf{x}_*) \right] \\ &\quad - \eta \mathbf{Z} \nabla f(\mathbf{x}_*), \quad (108)\end{aligned}$$

where in (a) we again used the MHT on $f(\cdot)$, and in (b) we used (107). Finally, collecting the terms in (108) gives

$$\begin{aligned}&\left(\mathbf{I} - \mathbf{W} + \eta \mathbf{W} \sum_{i=0}^{T-2} \mathbf{A}_i + \eta \mathbf{Z} \mathbf{A}_{T-1} + \mathcal{O}(\eta^2 T^2) \right) (\hat{\mathbf{x}} - \mathbf{x}_*) \\ &= \left(-\eta [(T-1) \mathbf{W} + \mathbf{Z}] + \eta^2 (T-1) \mathbf{Z} \mathbf{A}_{T-1} + \eta^2 \mathbf{W} \sum_{i=1}^{T-1} \sum_{j=i}^{T-2} \mathbf{A}_j + \mathcal{O}(\eta^3 T^3) \right) \nabla f(\mathbf{x}_*). \quad (109)\end{aligned}$$

We will now, based on (109), obtain a bound on $\|\hat{\mathbf{x}} - \mathbf{x}_*\|$, by writing

$$\hat{\mathbf{x}} - \mathbf{x}_* = \Pi_1^\perp (\hat{\mathbf{x}} - \mathbf{x}_*) + \Pi_1 (\hat{\mathbf{x}} - \mathbf{x}_*) \quad (110)$$

and separately considering the two terms $\|\Pi_1^\perp (\hat{\mathbf{x}} - \mathbf{x}_*)\|$ and $\|\Pi_1 (\hat{\mathbf{x}} - \mathbf{x}_*)\|$, and using that

$$\|\hat{\mathbf{x}} - \mathbf{x}_*\| \leq \|\Pi_1^\perp (\hat{\mathbf{x}} - \mathbf{x}_*)\| + \|\Pi_1 (\hat{\mathbf{x}} - \mathbf{x}_*)\|. \quad (111)$$

First, multiplication of (109) from the left by $(\mathbf{I} - \mathbf{W})^\dagger$, using the identity $(\mathbf{I} - \mathbf{W})^\dagger (\mathbf{I} - \mathbf{W}) = \Pi_1^\perp$, gives

$$\begin{aligned}&\left[\Pi_1^\perp + \eta (\mathbf{I} - \mathbf{W})^\dagger \left(\mathbf{W} \sum_{i=0}^{T-2} \mathbf{A}_i + \mathbf{Z} \mathbf{A}_{T-1} \right) + \mathcal{O}(\eta^2 T^2) \right] (\hat{\mathbf{x}} - \mathbf{x}_*) \\ &= \left[-\eta (\mathbf{I} - \mathbf{W})^\dagger [(T-1) \mathbf{W} + \mathbf{Z}] + \eta^2 (\mathbf{I} - \mathbf{W})^\dagger \left((T-1) \mathbf{Z} \mathbf{A}_{T-1} + \mathbf{W} \sum_{i=1}^{T-1} \sum_{j=i}^{T-2} \mathbf{A}_j \right) + \mathcal{O}(\eta^3 T^3) \right] \nabla f(\mathbf{x}_*). \quad (112)\end{aligned}$$

By taking the norm of both sides of (112) and using the triangle inequality, we obtain an upper bound on the first term in (111):

$$\|\Pi_1^\perp (\hat{\mathbf{x}} - \mathbf{x}_*)\| \leq \eta \|(\mathbf{I} - \mathbf{W})^\dagger [(T-1) \mathbf{W} + \mathbf{Z}]\|_2 \|\nabla f(\mathbf{x}_*)\| + \mathcal{O}(\eta T) \|\hat{\mathbf{x}} - \mathbf{x}_*\| + \mathcal{O}(\eta^2 T^2). \quad (113)$$

Next, multiplication of (109) from the left by $\mathbf{1}^T$ and using again (110), along with (34), and that $\mathbf{W}\mathbf{1} = \mathbf{Z}\mathbf{1} = \mathbf{1}$ gives

$$\mathbf{1}^T \left(\sum_{i=0}^{T-1} \mathbf{A}_i + \mathcal{O}(\eta T^2) \right) \Pi_1(\hat{\mathbf{x}} - \mathbf{x}_*) = -\mathbf{1}^T \left(\sum_{i=0}^{T-1} \mathbf{A}_i + \mathcal{O}(\eta T^2) \right) \Pi_1^\perp(\hat{\mathbf{x}} - \mathbf{x}_*) + \mathbf{1}^T \left[\eta \sum_{i=1}^{T-1} \sum_{j=i}^{T-1} \mathbf{A}_j + \mathcal{O}(\eta^2 T^3) \right] \nabla f(\mathbf{x}_*).$$

By using the properties of $\{\mathbf{A}_i\}$ we establish that

$$\begin{aligned} (\mu + \mathcal{O}(\eta T)) \|\Pi_1(\hat{\mathbf{x}} - \mathbf{x}_*)\| &= \frac{1}{\sqrt{N}} (\mu + \mathcal{O}(\eta T)) \|\mathbf{1}^T(\hat{\mathbf{x}} - \mathbf{x}_*)\| \\ &\leq \frac{1}{\sqrt{N}} \left\| \left(\frac{1}{TN} \sum_{i=0}^{T-1} \mathbf{1}^T \mathbf{A}_i \mathbf{1} + \mathcal{O}(\eta T) \right) \mathbf{1}^T(\hat{\mathbf{x}} - \mathbf{x}_*) \right\| \\ &= \frac{1}{T\sqrt{N}} \left\| \mathbf{1}^T \left(\sum_{i=0}^{T-1} \mathbf{A}_i + \mathcal{O}(\eta T^2) \right) \Pi_1(\hat{\mathbf{x}} - \mathbf{x}_*) \right\| \\ &= \frac{1}{T\sqrt{N}} \left\| -\mathbf{1}^T \left(\sum_{i=0}^{T-1} \mathbf{A}_i + \mathcal{O}(\eta T^2) \right) \Pi_1^\perp(\hat{\mathbf{x}} - \mathbf{x}_*) + \mathbf{1}^T \left(\eta \sum_{i=1}^{T-1} \sum_{j=i}^{T-1} \mathbf{A}_j + \mathcal{O}(\eta^2 T^3) \right) \nabla f(\mathbf{x}_*) \right\| \\ &\leq (L + \mathcal{O}(\eta T)) \|\Pi_1^\perp(\hat{\mathbf{x}} - \mathbf{x}_*)\| + \frac{1}{T} \left[\eta L \frac{T(T-1)}{2} + \mathcal{O}(\eta^2 T^3) \right] \|\nabla f(\mathbf{x}_*)\|, \end{aligned}$$

which gives the following upper bound on the second term in (111):⁶

$$\begin{aligned} \|\Pi_1(\hat{\mathbf{x}} - \mathbf{x}_*)\| &\leq \left(\frac{L}{\mu} + \mathcal{O}(\eta T) \right) \|\Pi_1^\perp(\hat{\mathbf{x}} - \mathbf{x}_*)\| + \eta \frac{(T-1)L}{2\mu} \|\nabla f(\mathbf{x}_*)\| + \mathcal{O}(\eta^2 T^2) \\ &\leq \eta \frac{L}{\mu} \left(\left\| (\mathbf{I} - \mathbf{W})^\dagger [(T-1)\mathbf{W} + \mathbf{Z}] \right\|_2 \|\nabla f(\mathbf{x}_*)\| + \frac{T-1}{2} \|\nabla f(\mathbf{x}_*)\| \right) \\ &\quad + \mathcal{O}(\eta T) \|\hat{\mathbf{x}} - \mathbf{x}_*\| + \mathcal{O}(\eta^2 T^2). \end{aligned} \tag{114}$$

Adding (113) and (114), using (111), gives

$$\|\hat{\mathbf{x}} - \mathbf{x}_*\| \leq \eta \left[\frac{L}{2\mu} (T-1) + \left(1 + \frac{L}{\mu} \right) \left\| (\mathbf{I} - \mathbf{W})^\dagger [(T-1)\mathbf{W} + \mathbf{Z}] \right\|_2 \right] \|\nabla f(\mathbf{x}_*)\| + \mathcal{O}(\eta T) \|\hat{\mathbf{x}} - \mathbf{x}_*\| + \mathcal{O}(\eta^2 T^2),$$

from which, finally,

$$\|\hat{\mathbf{x}} - \mathbf{x}_*\| \leq \eta \left[\frac{L}{2\mu} (T-1) + \left(1 + \frac{L}{\mu} \right) \left\| (\mathbf{I} - \mathbf{W})^\dagger [(T-1)\mathbf{W} + \mathbf{Z}] \right\|_2 \right] \|\nabla f(\mathbf{x}_*)\| + \mathcal{O}(\eta^2 T^2). \tag{115}$$

Explicit Bounds without $\mathcal{O}(\eta^2 T^2)$ Terms

Throughout the treatment of multiple local gradient updates, we give bounds on $\|\hat{\mathbf{x}} - \mathbf{x}_*\|$ that contain higher-order terms expressed in terms of $\mathcal{O}(\cdot)$. This choice was made only to maximize simplicity of the exposition, and the bounds with $\mathcal{O}(\cdot)$ are sufficient to show how $\|\hat{\mathbf{x}} - \mathbf{x}_*\|$ scales with different parameters of the problem.

But importantly, using the bound on $\|\mathbf{E}_i\|_2$ provided in (105), every $\mathcal{O}(\cdot)$ -term may be replaced by an explicit upper bound. In particular, let $\alpha > 1$ be arbitrary. If $g(\eta T) \leq h(\eta T) + \mathcal{O}(\eta^2 T^2)$ then there exists a constant $\eta_0 T_0$ such that $g(\eta T) \leq \alpha h(\eta T)$ when $\eta T \geq \eta_0 T_0$. While explicit expressions for $\eta_0 T_0$ can be obtained, they become rather intricate and do not add insight.

Additional Remark

The following observation, which follows from the calculation above, will be used in the analysis of the noisy case. By taking the norm on both sides of (107), we obtain $\|\tilde{\mathbf{x}}_{T-1} - \mathbf{x}_*\| = \|\hat{\mathbf{x}} - \mathbf{x}_*\| + \mathcal{O}(\eta T)$ and by a slight amendment of the calculation (re-doing the recursion up to i instead of $T-1$) that

$$\|\tilde{\mathbf{x}}_i - \mathbf{x}_*\| = \|\hat{\mathbf{x}} - \mathbf{x}_*\| + \mathcal{O}(\eta T), \quad i = 0, \dots, T-1.$$

⁶Note that if $[1 + \mathcal{O}(\eta)]x = a + \eta b + \mathcal{O}(\eta^2)$ then $x = (1 + \mathcal{O}(\eta))a + \eta b + \mathcal{O}(\eta^2)$.

Then for $i = 0, \dots, T-1$,

$$\|\tilde{\mathbf{x}}_i\| \leq \|\mathbf{x}_*\| + \|\tilde{\mathbf{x}}_i - \mathbf{x}_*\| \leq \|\mathbf{x}_*\| + \|\hat{\mathbf{x}} - \mathbf{x}_*\| + \mathcal{O}(\eta T) \stackrel{(a)}{=} \|\mathbf{x}_*\| + \mathcal{O}(\eta T), \quad (116)$$

where (a) follows from (115).

C. Gradient Noise in Stochastic GD (and DGD/Diffusion) Can Have Unbounded Variance

To see why gradient noise in stochastic GD (and DGD) may have unbounded variance, consider GD with linear regression in two dimensions. Suppose the parameter is the slope, x , of a straight line that passes through the origin, and suppose three data points are available: $(1, 1)$, $(1, -1)$, and $(2, 0)$. The objective is

$$f(x) = \frac{1}{3}[(1-x)^2 + (-1-x)^2 + (0-2x)^2],$$

each term corresponding to each respective data point. For $f(x)$, we have $f'(x) = 4x$ and $\mu = L = 4$, so with $\eta = 1/4$, GD on $f(x)$ would yield the optimal solution, $x = 0$, in a single step.

Now suppose stochastic GD is applied, sampling one of the data points (objectives) uniformly at random, and computing the derivative (gradient). This results in the noisy gradient,

$$\begin{cases} \frac{d}{dx}(1-x)^2 = 2x-2, & \text{with probability } 1/3 \\ \frac{d}{dx}(-1-x)^2 = 2x+2, & \text{with probability } 1/3 \\ \frac{d}{dx}(0-2x)^2 = 8x, & \text{with probability } 1/3 \end{cases} \\ = 4x + \epsilon,$$

where ϵ is the gradient noise. This noise has the distribution (conditional on x)

$$\epsilon = \begin{cases} -2x-2, & \text{with probability } 1/3 \\ 2-2x, & \text{with probability } 1/3, \\ 4x, & \text{with probability } 1/3 \end{cases}$$

and the moments

$$\begin{aligned} \mathbb{E}\{\epsilon|x\} &= 0, \\ \mathbb{E}\{\epsilon^2|x\} &= \frac{8}{3} + 8x^2. \end{aligned} \quad (117)$$

The second moment in (117) has the form in (14), and is unbounded.

Linear-regression examples in arbitrary dimensions, and with arbitrary numbers of data points, can be similarly constructed.

An example of a model that results in bounded gradient noise variance is when the objective is sampled from a family of functions whose gradients are uniformly bounded.⁷ However, this model is incompatible with the assumption on strong convexity of the objective, since strongly convex functions cannot have uniformly bounded gradients.

An example of a model with a strongly convex objective that results in bounded gradient noise variance is when the objective is sampled from a family of convex (but not strictly convex) functions with uniformly bounded gradients, to which a deterministic, strongly convex regularization term is added.

⁷For differentiable objective functions, the gradient is uniformly bounded if and only if the function is Lipschitz continuous. Such Lipschitz continuity of the objective should not be conflated with L -smoothness, which is equivalent to Lipschitz-continuity of the *gradient*.